

TradingAI Database Architecture, Schema, Tables,

Complete current source-defined database catalog through m78_001.

System reference date: 2026-08-29

Standalone current-state reference for the complete platform through Milestone 78.

1. Current database architecture

TradingAI uses PostgreSQL 17 as its authoritative relational store. SQLAlchemy 2.x provides model/repository access and Alembic provides versioned schema evolution. The database holds raw/normalized observations, intelligence snapshots, publications, lineage, portfolio/risk state, trade plans, execution/broker truth, performance evidence, operations state and isolated research/shadow evidence.

The current source-defined inventory is reconstructed from the TradingPlatform_new source plus the additive Milestone 78 migration. The base archive contains 201 named tables; m78_001 adds 8 Setup Intelligence tables, producing a current code-defined inventory of 209 named tables. Because a live pg_catalog dump was not part of the attached source package, environment-specific runtime drift is outside this catalog.

2. Schema families and authority roles

Family	Representative authority
market / sector / correlation / sentiment / dealer	Market Overview and market-intelligence snapshots.
trend / stock / scanner / lineage	Multi-timeframe trend and current Stock Intelligence authority.
institutional / option / inflection / valuation	Institutional Options, Decision Intelligence, inflection and valuation evidence.
portfolio / risk / allocation / optimization	Portfolio snapshots, risk budgets, optimization and current allocation publications.
advanced trade plan / execution / broker	Trade Builder, canonical order intent, working-order lifecycle, fills and broker truth.
performance / outcome	Prediction-vs-actual, calibration, execution/management-quality evidence.
production / live / operations	Freshness, health, run control, recovery and live-governance records.
historical / m77	Historical replay and governed research/prospective evidence.
setup_intelligence / setup_probability	M78 shadow-only setup snapshots, outcomes, models, predictions, publications and certifications.

3. Milestone 78 schema

Table	Current columns and constraints
setup_intelligence_snapshots	setup_id PK; candidate_id; scanner_run_id; symbol; as_of; setup_type; setup_family; stage; direction; quality; confidence; invalidation_level; entry_reference; source_state_hash; context_json; evidence_json; lineage_json; authority_effect default 0; state_hash; captured_at. UNIQUE(candidate_id, setup_type, as_of). Indexed on candidate/scanner/symbol/as_of/setup/family/stage/direction/hash/time fields.
setup_intelligence_transitions	transition_id PK; setup_id; symbol; setup_type; from_stage; to_stage; occurred_at; reason; payload_json. Indexed on setup, symbol, setup type, stages and time.
setup_intelligence_outcomes	observation_id PK; setup_id; candidate_id; symbol; as_of; setup_type; stage; direction; market_regime; gamma_regime; sector_regime; volatility_regime; label_version; status; target_1/2/3_before_stop; thesis_invalidations; profitable_at_horizon; maximum_favorable_excursion_pct; maximum_adverse_excursion_pct; realized_return_pct; days_to_target_1; days_to_stop; context_json; label_json; materialized_at. UNIQUE(setup_id, label_version).
setup_probability_model_artifacts	model_id PK; model_version UNIQUE; lifecycle_state; sample_size; training_cutoff; artifact_json; evaluation_json; governance_json; state_hash; created_at; approved_by/approved_at; activated_by/activated_at.
setup_probability_predictions	prediction_id PK; setup_id; symbol; model_id; predicted_at; status; target_1_probability; profitable_probability; expected_return_pct; expected_r; uncertainty; assessment_json. UNIQUE(setup_id, model_id).
setup_intelligence_publications	publication_id PK; publication_name; source_scanner_run_id; status; setup_count; published_at; payload_json; authority_effect default 0.
setup_intelligence_certifications	certification_id PK; setup_type; model_id; state; historical_gate; prospective_gate; evidence_json; authority_effect default 0; certified_by; certified_at.
setup_intelligence_audit_events	event_id PK; entity_id; event_type; actor; reason; occurred_at; payload_json.

4. Complete base schema catalog

The following 201 table definitions come from the current TradingPlatform_new source archive. M78 tables are documented above and should be considered part of the same current database.

1. advanced_trade_plan_audit (8 columns)

Migration: 20260730_m56_advanced_trade_builder.py | Model: src/trading_ai/advanced_trade_builder/models.py

Column	Type	Null / PK / Unique / FK-default metadata
audit_id	sa.String(128)	None True False
trade_plan_id	sa.String(128)	False False False
trade_plan_version	sa.Integer()	False False False
event_type	sa.String(64)	False False False
actor	sa.String(128)	False False False
reason	sa.Text()	False False False
event_timestamp	sa.String(64)	False False False
payload_json	sa.JSON()	False False False

Indexes: op.create_index('ix_m56_trade_audit_plan', 'advanced_trade_plan_audit', ['trade_plan_id'])

2. advanced_trade_plans (26 columns)

Migration: 20260730_m56_advanced_trade_builder.py | Model: src/trading_ai/advanced_trade_builder/models.py

Column	Type	Null / PK / Unique / FK-default metadata
trade_plan_id	sa.String(128)	None True False
opportunity_id	sa.String(128)	False False False
opportunity_version	sa.Integer()	False False False
intelligence_id	sa.String(128)	None False False
account_id	sa.String(128)	False False False

Column	Type	Null / PK / Unique / FK-default metadata
symbol	sa.String(32)	False False False
direction	sa.String(16)	False False False
strategy	sa.String(64)	False False False
state	sa.String(32)	False False False
version	sa.Integer()	False False False
capital	sa.Float()	False False False
risk_budget_pct	sa.Float()	False False False
risk_budget_amount	sa.Float()	False False False
estimated_debit	sa.Float()	False False False
estimated_credit	sa.Float()	False False False
max_loss	sa.Float()	False False False
max_profit	sa.Float()	None False False
reward_risk_ratio	sa.Float()	None False False
net_greeks_json	sa.JSON()	False False False
validation_json	sa.JSON()	False False False
legs_json	sa.JSON()	False False False
execution_intent_json	sa.JSON()	False False False
notes	sa.Text()	False False False
created_by	sa.String(128)	False False False
created_at	sa.String(64)	False False False
updated_at	sa.String(64)	False False False

Indexes: op.create_index('ix_m56_trade_plan_opportunity', 'advanced_trade_plans', ['opportunity_id'])

3. broker_account_bindings (15 columns)

Migration: m50_001_ibkr_paper_account_foundation.py | Model: src/trading_ai/broker/ibkr/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
binding_id	sa.String(128)	None True False
portfolio_id	sa.String(64)	False False False
broker_name	sa.String(64)	False False False
broker_environment	sa.String(16)	False False False
broker_account_id	sa.String(64)	False False False
base_currency	sa.String(8)	False False False
host	sa.String(255)	False False False
port	sa.Integer()	False False False
client_id	sa.Integer()	False False False
read_only	sa.Boolean()	False False False server_default=sa.true()
live_trading_enabled	sa.Boolean()	False False False server_default=sa.false()
status	sa.String(32)	False False False
created_at	sa.String(64)	False False False
updated_at	sa.String(64)	False False False
metadata_json	sa.JSON()	False False False

Indexes: op.create_index('ix_broker_account_bindings_portfolio_id', 'broker_account_bindings', ['portfolio_id'])

4. broker_account_snapshots (12 columns)

Migration: m50_001_ibkr_paper_account_foundation.py | Model: src/trading_ai/broker/ibkr/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_id	sa.String(128)	None True False
binding_id	sa.String(128)	False False False
portfolio_id	sa.String(64)	False False False
broker_account_id	sa.String(64)	False False False
captured_at	sa.String(64)	False False False
base_currency	sa.String(8)	False False False
net_liquidation	sa.Float()	False False False
total_cash_value	sa.Float()	False False False
available_funds	sa.Float()	False False False
buying_power	sa.Float()	False False False
excess_liquidity	sa.Float()	False False False server_default=0
raw_json	sa.JSON()	False False False

Indexes: op.create_index(fix_broker_account_snapshots_{column}', 'broker_account_snapshots', [column])

5. broker_current_positions (30 columns)

Migration: m63_001_broker_portfolio_sync.py | Model: src/trading_ai/broker_portfolio_sync/models.py

Column	Type	Null / PK / Unique / FK-default metadata
broker_position_id	sa.String(160)	None True False
portfolio_id	sa.String(64)	False False False
binding_id	sa.String(128)	False False False
broker_account_id	sa.String(64)	False False False

Indexes: op.create_index(fix_m63_current_{col}', 'broker_current_positions', [col])

6. broker_executions (21 columns)

Migration: m50_002_ibkr_paper_order_routing.py | Model: src/trading_ai/broker/ibkr/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
execution_id	sa.String(128)	None True False
broker_order_record_id	sa.String(160)	False False False
binding_id	sa.String(128)	False False False
portfolio_id	sa.String(64)	False False False
aggregate_id	sa.String(128)	False False False
broker_account_id	sa.String(64)	False False False
broker_order_id	sa.Integer()	False False False
permanent_id	sa.Integer()	False False False server_default=0
contract_id	sa.Integer()	False False False server_default=0
symbol	sa.String(64)	False False False
security_type	sa.String(16)	False False False
side	sa.String(16)	False False False
quantity	sa.Float()	False False False
price	sa.Float()	False False False
commission	sa.Float()	False False False server_default=0
currency	sa.String(8)	False False False server_default=USD
exchange	sa.String(32)	False False False server_default=
executed_at	sa.String(64)	False False False

Column	Type	Null / PK / Unique / FK-default metadata
imported_at	sa.String(64)	False False False

7. broker_order_controls (10 columns)

Migration: m50_002_ibkr_paper_order_routing.py | Model: src/trading_ai/broker/ibkr/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
portfolio_id	sa.String(64)	None True False
paper_order_submission_	sa.Boolean()	False False False server_default=sa.false()
activation_token_hash	sa.String(128)	False False False server_default=
activated_at	sa.String(64)	None False False
activated_by	sa.String(128)	False False False server_default=
disabled_at	sa.String(64)	None False False
disable_reason	sa.Text()	False False False server_default=
version	sa.Integer()	False False False server_default=1
updated_at	sa.String(64)	False False False
metadata_json	sa.JSON()	False False False server_default=sa.text("{}::json")

8. broker_orders (25 columns)

Migration: m50_002_ibkr_paper_order_routing.py | Model: src/trading_ai/broker/ibkr/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
broker_order_record_id	sa.String(160)	None True False
binding_id	sa.String(128)	False False False
portfolio_id	sa.String(64)	False False False
aggregate_id	sa.String(128)	False False False
client_order_id	sa.String(128)	False False False
broker_account_id	sa.String(64)	False False False
broker_order_id	sa.Integer()	False False False
permanent_id	sa.Integer()	False False False server_default=0
api_client_id	sa.Integer()	False False False
symbol	sa.String(64)	False False False
security_type	sa.String(16)	False False False
side	sa.String(16)	False False False
quantity	sa.Float()	False False False
order_type	sa.String(16)	False False False
time_in_force	sa.String(16)	False False False
limit_price	sa.Float()	None False False
stop_price	sa.Float()	None False False
status	sa.String(32)	False False False
filled_quantity	sa.Float()	False False False server_default=0
remaining_quantity	sa.Float()	False False False
average_fill_price	sa.Float()	False False False server_default=0
submitted_at	sa.String(64)	False False False
updated_at	sa.String(64)	False False False
last_error	sa.Text()	False False False server_default=
raw_json	sa.JSON()	False False False server_default=sa.text("{}::json")

9. broker_portfolio_alerts (10 columns)

Migration: m63_001_broker_portfolio_sync.py | Model: src/trading_ai/broker_portfolio_sync/models.py

Column	Type	Null / PK / Unique / FK-default metadata
alert_id	sa.String(128)	None True False
portfolio_id	sa.String(64)	False False False
broker_position_id	sa.String(160)	None False False
severity	sa.String(16)	False False False
alert_type	sa.String(64)	False False False
status	sa.String(32)	False False False server_default=OPEN
message	sa.Text()	False False False
created_at	sa.String(64)	False False False
resolved_at	sa.String(64)	None False False
payload_json	postgresql.JSONB(astext_type=sa.Text)	False False

Indexes: op.create_index('ix_m63_alert_{col}', 'broker_portfolio_alerts', [col])

10. broker_portfolio_publications (13 columns)

Migration: m63_001_broker_portfolio_sync.py | Model: src/trading_ai/broker_portfolio_sync/models.py

Column	Type	Null / PK / Unique / FK-default metadata
publication_id	sa.String(128)	None True False
publication_name	sa.String(128)	False False False
portfolio_id	sa.String(64)	False False False
broker_account_id	sa.String(64)	False False False
account_snapshot_id	sa.String(128)	False False False
reconciliation_run_id	sa.String(128)	False False False
status	sa.String(32)	False False False
position_count	sa.Integer()	False False False
matched_count	sa.Integer()	False False False server_default=0
broker_discovered_count	sa.Integer()	False False False server_default=0
drift_count	sa.Integer()	False False False server_default=0
published_at	sa.String(64)	False False False
payload_json	postgresql.JSONB(astext_type=sa.Text)	False False

Indexes: op.create_index('ix_m63_publication_{col}', 'broker_portfolio_publications', [col])

11. broker_position_snapshots (18 columns)

Migration: m50_001_ibkr_paper_account_foundation.py | Model: src/trading_ai/broker_ibkr/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_position_id	sa.String(160)	None True False
account_snapshot_id	sa.String(128)	False False False
portfolio_id	sa.String(64)	False False False
broker_account_id	sa.String(64)	False False False
contract_id	sa.Integer()	False False False
symbol	sa.String(32)	False False False
local_symbol	sa.String(64)	False False False server_default=
security_type	sa.String(16)	False False False
currency	sa.String(8)	False False False

Column	Type	Null / PK / Unique / FK-default metadata
exchange	sa.String(32)	False False False
quantity	sa.Float()	False False False
average_cost	sa.Float()	False False False
expiry	sa.String(16)	False False False server_default=
strike	sa.Float()	None False False
right	sa.String(8)	False False False server_default=
multiplier	sa.Float()	False False False server_default=1

Indexes: op.create_index('ix_broker_position_snapshots_{column}', 'broker_position_snapshots', [column])

12. broker_reconciliation_runs (10 columns)

Migration: m50_001_ibkr_paper_account_foundation.py | Model: src/trading_ai/broker/ibkr/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
run_id	sa.String(128)	None True False
binding_id	sa.String(128)	False False False
portfolio_id	sa.String(64)	False False False
started_at	sa.String(64)	False False False
completed_at	sa.String(64)	False False False
status	sa.String(32)	False False False
account_match	sa.Boolean()	False False False
details_json	sa.JSON()	False False False
error_text	sa.Text()	False False False server_default=

Indexes: op.create_index('ix_broker_reconciliation_runs_binding_id', 'broker_reconciliation_runs', ['binding_id'])

13. canonical_order_events (17 columns)

Migration: m49_001_authoritative_paper_trading.py | Model: src/trading_ai/authoritative_paper_trading/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
event_id	sa.String(128)	None True False
event_type	sa.String(64)	False False False
aggregate_id	sa.String(128)	False False False
aggregate_version	sa.Integer()	False False False
client_order_id	sa.String(128)	False False False
account_id	sa.String(64)	False False False
event_timestamp	sa.String(64)	False False False
previous_state	sa.String(32)	False False False
new_state	sa.String(32)	False False False
broker_order_id	sa.String(128)	None False False
filled_quantity	sa.Float()	False False False server_default=0
remaining_quantity	sa.Float()	False False False server_default=0
average_fill_price	sa.Float()	None False False
reason	sa.Text()	None False False
correlation_id	sa.String(128)	None False False
causation_id	sa.String(128)	None False False
metadata_json	sa.JSON()	False False False

Indexes: op.create_index(name, 'canonical_order_events', cols)

14. canonical_orders (26 columns)

Migration: m49_001_authoritative_paper_trading.py | Model: src/trading_ai/authoritative_paper_trading/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
aggregate_id	sa.String(128)	None True False
client_order_id	sa.String(128)	False False True
account_id	sa.String(64)	False False False
idempotency_key	sa.String(160)	False False True
order_type	sa.String(32)	False False False
time_in_force	sa.String(32)	False False False
state	sa.String(32)	False False False
version	sa.Integer()	False False False
total_quantity	sa.Float()	False False False
filled_quantity	sa.Float()	False False False server_default=0
remaining_quantity	sa.Float()	False False False
average_fill_price	sa.Float()	None False False
limit_price	sa.Float()	None False False
stop_price	sa.Float()	None False False
outside_regular_hours	sa.Boolean()	False False False server_default=sa.false()
strategy_name	sa.String(128)	None False False
broker_order_id	sa.String(128)	None False False
parent_aggregate_id	sa.String(128)	None False False
root_aggregate_id	sa.String(128)	None False False
replace_count	sa.Integer()	False False False server_default=0
legs_json	sa.JSON()	False False False
created_at	sa.String(64)	False False False
updated_at	sa.String(64)	False False False
terminal_at	sa.String(64)	None False False
last_event_id	sa.String(128)	None False False
metadata_json	sa.JSON()	False False False

Indexes: op.create_index(name, 'canonical_orders', cols)

15. correlation_pair_snapshot (8 columns)

Migration: m46_001_market_intelligence.py | Model: src/trading_ai/market_intelligence/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_timestamp	sa.DateTime(timezone=True)	None True False
symbol_a	sa.String(16)	None True False
symbol_b	sa.String(16)	None True False
as_of_date	sa.Date()	False False False
lookback_days	sa.Integer()	False False False
correlation	sa.Float()	False False False
confidence	sa.Float()	False False False
provenance	sa.String(32)	False False False

16. correlation_snapshot (12 columns)

Migration: m46_001_market_intelligence.py | Model: src/trading_ai/market_intelligence/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_timestamp	sa.DateTime(timezone=True)	None True False
universe_name	sa.String(64)	None True False
as_of_date	sa.Date()	False False False
lookback_days	sa.Integer()	False False False
dispersion	sa.Float()	False False False
correlation_regime	sa.String(32)	False False False

17. dealer_expiration_profile (4 columns)

Migration: m44_002_normalized_market_structure.py | Model: src/trading_ai/institutional_market_structure/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
symbol	sa.String(16)	None True False
as_of_date	sa.Date()	None True False
expiry	sa.Date()	None True False
dte	sa.Integer()	False False False

18. dealer_position_change_snapshot (8 columns)

Migration: m46_001_market_intelligence.py | Model: src/trading_ai/market_intelligence/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_timestamp	sa.DateTime(timezone=True)	None True False
symbol	sa.String(16)	None True False
as_of_date	sa.Date()	False False False
positioning_score	sa.Float()	False False False
dealer_conviction_score	sa.Float()	False False False
confidence	sa.Float()	False False False
provenance	sa.String(32)	False False False
payload_json	sa.Text()	False False False

19. dealer_position_snapshot (47 columns)

Migration: m44_002_normalized_market_structure.py | Model: src/trading_ai/institutional_market_structure/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
symbol	sa.String(16)	None True False
as_of_date	sa.Date()	None True False
quote_date	sa.Date()	False False False
spot_price	sa.Float()	False False False
source_table	sa.String(128)	False False False
dealer_sign_convention	sa.String(32)	False False False
estimator_name	sa.String(96)	False False False
estimator_version	sa.String(24)	False False False
source_contract_count	sa.Integer()	False False False
executable_contract_cou	sa.Integer()	False False False
quote_coverage_pct	sa.Float()	False False False
net_gamma_exposure	sa.Float()	False False False
net_delta_exposure	sa.Float()	False False False
net_vanna_exposure	sa.Float()	False False False

Column	Type	Null / PK / Unique / FK-default metadata
net_charm_exposure	sa.Float()	False False False
gamma_regime	sa.String(32)	False False False
gamma_flip	sa.Float()	None False False
gamma_flip_distance_pct	sa.Float()	None False False
gamma_flip_confidence	sa.Float()	False False False
primary_call_wall	sa.Float()	None False False
secondary_call_wall	sa.Float()	None False False

20. dealer_strike_profile (7 columns)

Migration: m44_002_normalized_market_structure.py | Model: src/trading_ai/institutional_market_structure/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
symbol	sa.String(16)	None True False
as_of_date	sa.Date()	None True False
expiry	sa.Date()	None True False
strike	sa.Float()	None True False
dte	sa.Integer()	False False False
trade_eligible	sa.Boolean()	False False False

21. execution_fill_events (12 columns)

Migration: m70_003_execution_intelligence_completion.py | Model: src/trading_ai/execution_intelligence/models.py

Column	Type	Null / PK / Unique / FK-default metadata
fill_event_id	sa.String(160)	None True False
execution_intent_id	sa.String(128)	False False False
aggregate_id	sa.String(128)	False False False
broker_execution_id	sa.String(128)	False False True
broker_order_id	sa.Integer()	False False False
symbol	sa.String(64)	False False False
side	sa.String(16)	False False False
quantity	sa.Float()	False False False
price	sa.Float()	False False False
commission	sa.Float()	False False False server_default=0
executed_at	sa.String(64)	False False False
payload_json	postgresql.JSONB(astext_type=sa.Text)	False False server_default=sa.text("{}::jsonb")

Indexes: op.create_index('ix_m70_fills_intent', 'execution_fill_events', ['execution_intent_id', 'executed_at'])

22. execution_intelligence_events (8 columns)

Migration: m70_001_institutional_execution_intelligence.py | Model: src/trading_ai/execution_intelligence/models.py

Column	Type	Null / PK / Unique / FK-default metadata
event_id	sa.String(128)	None True False
execution_snapshot_id	sa.String(128)	False False False
execution_intent_id	sa.String(128)	False False False
event_type	sa.String(64)	False False False
actor	sa.String(128)	False False False
reason	sa.Text()	False False False

Column	Type	Null / PK / Unique / FK-default metadata
payload_json	postgresql.JSONB(astext_type=sa.Text False	False False
created_at	sa.String(64)	False False False

Indexes: op.create_index('ix_m70_event_intent_created', 'execution_intelligence_events', ['execution_intent_id', 'created_at'])

23. execution_intelligence_snapshots (21 columns)

Migration: m70_001_institutional_execution_intelligence.py | Model: src/trading_ai/execution_intelligence/models.py

Column	Type	Null / PK / Unique / FK-default metadata
execution_snapshot_id	sa.String(128)	None True False
execution_intent_id	sa.String(128)	False False False
execution_intent_version	sa.Integer()	False False False
trade_plan_id	sa.String(128)	False False False
symbol	sa.String(32)	False False False
strategy	sa.String(64)	False False False
decision	sa.String(32)	False False False
execution_confidence	sa.Float()	False False False
fresh_executable_price	sa.Float()	False False False
governed_limit_price	sa.Float()	False False False
adverse_price_drift_pct	sa.Float()	False False False
quote_age_seconds	sa.Float()	False False False
fresh_max_loss	sa.Float()	False False False
risk_budget_amount	sa.Float()	False False False
validation_json	postgresql.JSONB(astext_type=sa.Text False	False False
quotes_json	postgresql.JSONB(astext_type=sa.Text False	False False
envelope_json	postgresql.JSONB(astext_type=sa.Text False	False False
policy_json	postgresql.JSONB(astext_type=sa.Text False	False False
evidence_json	postgresql.JSONB(astext_type=sa.Text False	False False
created_at	sa.String(64)	False False False

Indexes: op.create_index('ix_m70_exec_intent_created', 'execution_intelligence_snapshots', ['execution_intent_id', 'created_at'])

24. execution_intent_audit_events (10 columns)

Migration: 20260803_m59_execution_workspace.py | Model: src/trading_ai/execution_workspace/models.py

Column	Type	Null / PK / Unique / FK-default metadata
event_id	sa.String(128)	None True False
execution_intent_id	sa.String(128)	False False False
execution_intent_version	sa.Integer	False False False
event_type	sa.String(64)	False False False
previous_state	sa.String(32)	None False False
new_state	sa.String(32)	False False False
actor	sa.String(128)	False False False
reason	sa.Text	False False False
event_timestamp	sa.String(64)	False False False
payload_json	sa.JSON	False False False

25. execution_intents (24 columns)

Column	Type	Null / PK / Unique / FK-default metadata
execution_intent_id	sa.String(128)	None True False
trade_plan_id	sa.String(128)	False False False
trade_plan_version	sa.Integer	False False False
opportunity_id	sa.String(128)	False False False
portfolio_id	sa.String(64)	False False False
account_id	sa.String(128)	False False False
symbol	sa.String(32)	False False False
strategy	sa.String(64)	False False False
state	sa.String(32)	False False False
version	sa.Integer	False False False
max_loss	sa.Float	False False False
legs_json	sa.JSON	False False False
order_request_json	sa.JSON	False False False
validation_json	sa.JSON	False False False
broker_json	sa.JSON	False False False
metadata_json	sa.JSON	False False False
created_by	sa.String(128)	False False False
created_at	sa.String(64)	False False False
updated_at	sa.String(64)	False False False
submitted_at	sa.String(64)	None False False
terminal_at	sa.String(64)	None False False
execution_attempt	sa.Integer()	False False False server_default=1
retry_reason	sa.Text()	True False False

Indexes: op.create_index('ix_execution_intents_execution_attempt', 'execution_intents', ['execution_attempt'], unique=False)

26. execution_learning_samples (9 columns)

Migration: m70_003_execution_intelligence_completion.py | Model: src/trading_ai/execution_intelligence/models.py

Column	Type	Null / PK / Unique / FK-default metadata
execution_intent_id	sa.String(128)	None True False
symbol	sa.String(32)	False False False
strategy	sa.String(64)	False False False
outcome	sa.String(32)	False False False
execution_quality_score	sa.Float()	False False False
features_json	postgresql.JSONB(astext_type=sa.Text)	False False
outcome_json	postgresql.JSONB(astext_type=sa.Text)	False False
created_at	sa.String(64)	False False False
updated_at	sa.String(64)	False False False

Indexes: op.create_index('ix_m70_learning_outcome', 'execution_learning_samples', ['outcome'])

27. execution_orchestration_events (11 columns)

Migration: m38exec_execution_orchestration.py | Model: src/trading_ai/execution_orchestration/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
event_id	sa.String(128)	None True False
execution_order_id	sa.String(128)	False False False

Column	Type	Null / PK / Unique / FK-default metadata
event_type	sa.String(64)	False False False
from_status	sa.String(32)	False False False
to_status	sa.String(32)	False False False
occurred_at	sa.String(64)	False False False
details_json	sa.JSON()	False False False
JSON	?	False False False

Indexes: op.create_index('ix_exec_events_order_id', 'execution_orchestration_events', ['execution_order_id'])

28. execution_orchestration_orders (16 columns)

Migration: m38exec_execution_orchestration.py | Model: src/trading_ai/execution_orchestration/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
execution_order_id	sa.String(128)	None True False
client_order_id	sa.String(128)	False False False
portfolio_id	sa.String(64)	False False False
symbol	sa.String(32)	False False False
strategy	sa.String(64)	False False False
status	sa.String(32)	False False False
risk_status	sa.String(32)	False False False
approval_status	sa.String(32)	False False False
capital_limit	sa.Float()	False False False
broker_order_id	sa.String(128)	None False False
payload_json	sa.JSON()	False False False

Indexes: op.create_index('ix_exec_orders_client_order_id', 'execution_orchestration_orders', ['client_order_id'])

29. execution_orchestration_runs (11 columns)

Migration: m38exec_execution_orchestration.py | Model: src/trading_ai/execution_orchestration/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
run_id	sa.String(128)	None True False
status	sa.String(32)	False False False
trading_control	sa.String(64)	False False False
generated_at	sa.String(64)	False False False
payload_json	sa.JSON()	False False False
notes	sa.Text()	None False False
JSON	?	False False False
Text	?	None False False

Indexes: op.create_index('ix_exec_runs_status', 'execution_orchestration_runs', ['status'])

30. execution_order_telemetry (29 columns)

Migration: m70_003_execution_intelligence_completion.py | Model: src/trading_ai/execution_intelligence/models.py

Column	Type	Null / PK / Unique / FK-default metadata
execution_intent_id	sa.String(128)	None True False
trade_plan_id	sa.String(128)	False False False
execution_snapshot_id	sa.String(128)	None False False
aggregate_id	sa.String(128)	None False False

Column	Type	Null / PK / Unique / FK-default metadata
broker_order_id	sa.Integer()	None False False
symbol	sa.String(32)	False False False
strategy	sa.String(64)	False False False
state	sa.String(32)	False False False
fresh_midpoint_price	sa.Float()	False False False server_default=0
fresh_executable_price	sa.Float()	False False False server_default=0
submitted_limit_price	sa.Float()	False False False server_default=0
average_fill_price	sa.Float()	None False False
filled_quantity	sa.Float()	False False False server_default=0
remaining_quantity	sa.Float()	False False False server_default=0
commission_total	sa.Float()	False False False server_default=0
quote_age_seconds	sa.Float()	None False False
execution_confidence	sa.Float()	False False False server_default=0
market_drift_pct	sa.Float()	False False False server_default=0
execution_slippage_pct	sa.Float()	False False False server_default=0
realized_slippage_pct	sa.Float()	None False False
fill_rate_pct	sa.Float()	False False False server_default=0

Indexes: op.create_index('ix_m70_telemetry_state', 'execution_order_telemetry', ['state'])

31. execution_working_order_assessments (12 columns)

Migration: m70_003_execution_intelligence_completion.py | Model: src/trading_ai/execution_intelligence/models.py

Column	Type	Null / PK / Unique / FK-default metadata
assessment_id	sa.String(128)	None True False
execution_intent_id	sa.String(128)	False False False
execution_snapshot_id	sa.String(128)	False False False
state	sa.String(32)	False False False
recommended_action	sa.String(32)	False False False
current_limit_price	sa.Float()	False False False
governed_limit_price	sa.Float()	False False False
filled_quantity	sa.Float()	False False False server_default=0
remaining_quantity	sa.Float()	False False False server_default=0
working_age_seconds	sa.Float()	False False False server_default=0
evidence_json	postgresql.JSONB(astext_type=sa.Text)	False False server_default=sa.text("{}::jsonb")
created_at	sa.String(64)	False False False

Indexes: op.create_index('ix_m70_working_intent', 'execution_working_order_assessments', ['execution_intent_id', 'created_at'])

32. futures_bars (16 columns)

Migration: m71_002_futures_intelligence.py | Model: src/trading_ai/futures_intelligence/models.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.String(128)	None True False
product_code	sa.String(16)	False False False
ticker	sa.String(32)	False False False
resolution	sa.String(16)	False False False
window_start_ns	sa.String(32)	False False False

Column	Type	Null / PK / Unique / FK-default metadata
session_end_date	sa.String(16)	None False False
open	sa.Float()	False False False
high	sa.Float()	False False False
low	sa.Float()	False False False
close	sa.Float()	False False False
volume	sa.Float()	False False False
dollar_volume	sa.Float()	False False False
transactions	sa.Integer()	False False False
settlement_price	sa.Float()	None False False
source	sa.String(32)	False False False

Indexes: op.create_index(n, 'futures_bars', [c])

33. futures_contract_registry (10 columns)

Migration: m71_002_futures_intelligence.py | Model: src/trading_ai/futures_intelligence/models.py

Column	Type	Null / PK / Unique / FK-default metadata
ticker	sa.String(32)	None True False
product_code	sa.String(16)	False False False
as_of_date	sa.String(16)	False False False
first_trade_date	sa.String(16)	None False False
last_trade_date	sa.String(16)	None False False
settlement_date	sa.String(16)	None False False
days_to_maturity	sa.Integer()	None False False
trading_venue	sa.String(16)	None False False
active	sa.Integer()	False False False server_default=1
payload_json	sa.JSON()	False False False

Indexes: op.create_index('ix_futures_contract_product', 'futures_contract_registry', ['product_code'])

34. futures_intelligence_snapshots (17 columns)

Migration: m71_002_futures_intelligence.py | Model: src/trading_ai/futures_intelligence/models.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_id	sa.String(128)	None True False
product_code	sa.String(16)	False False False
ticker	sa.String(32)	False False False
snapshot_timestamp	sa.String(64)	False False False
last_price	sa.Float()	False False False
vwap	sa.Float()	None False False
overnight_high	sa.Float()	None False False
overnight_low	sa.Float()	None False False
rth_high	sa.Float()	None False False
rth_low	sa.Float()	None False False
trend_score	sa.Float()	False False False
momentum_score	sa.Float()	False False False
realized_volatility	sa.Float()	False False False
basis_pct	sa.Float()	None False False

Column	Type	Null / PK / Unique / FK-default metadata
confirmation_score	sa.Float()	False False False
state	sa.String(32)	False False False
payload_json	sa.JSON()	False False False

Indexes: op.create_index(n, 'futures_intelligence_snapshots', [c])

35. historical_replay_comparison (8 columns)

Migration: m47_002_historical_replay.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.BigInteger()	None True False

Indexes: op.create_index('ix_historical_replay_comparison_run', 'historical_replay_comparison', ['replay_run_id'])

36. historical_replay_run (19 columns)

Migration: m47_002_historical_replay.py

Column	Type	Null / PK / Unique / FK-default metadata
replay_run_id	sa.String(96)	None True False
replay_mode	sa.String(24)	False False False
source_scanner_run_id	sa.String(96)	True False False
source_decision_run_id	sa.String(96)	True False False
publication_name	sa.String(128)	True False False
ingestion_run_id	sa.String(128)	True False False
option_snapshot_id	sa.String(160)	True False False
market_state_hash	sa.String(64)	True False False
scanner_version	sa.String(64)	True False False
decision_engine_version	sa.String(64)	True False False
policy_version	sa.String(64)	True False False
started_at	sa.DateTime(timezone=True)	False False False
completed_at	sa.DateTime(timezone=True)	False False False
status	sa.String(24)	False False False
candidate_count	sa.Integer()	False False False server_default=0
decision_count	sa.Integer()	False False False server_default=0
mismatch_count	sa.Integer()	False False False server_default=0
warnings_json	sa.Text()	False False False server_default=[]
metadata_json	sa.Text()	False False False server_default={}

Indexes: op.create_index('ix_historical_replay_source_scanner', 'historical_replay_run', ['source_scanner_run_id'])

37. historical_underlying_replay_authority (11 columns)

Migration: m77_002_historical_underlying_replay_authority.py

Column	Type	Null / PK / Unique / FK-default metadata
symbol	sa.String(32)	None True False
disposition	sa.String(64)	False False False
continuity_start	sa.Date	False False False
eligible_from	sa.Date	None False False
last_date	sa.Date	False False False
valid_observation_count	sa.Integer	False False False

Column	Type	Null / PK / Unique / FK-default metadata
identity_break_count	sa.Integer	False False False
invalid_session_count	sa.Integer	False False False
authority_version	sa.String(96)	False False False
evidence_json	postgresql.JSONB	False False False
materialized_at	sa.DateTime(timezone=True)	False False False

Indexes: op.create_index('ix_m77_1_authority_disposition', 'historical_underlying_replay_authority', ['disposition'])

38. historical_underlying_replay_failure (7 columns)

Migration: m77_002_historical_underlying_replay_authority.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.BigInteger	None True False
replay_run_id	sa.String(128)	False False False
symbol	sa.String(32)	False False False
as_of	sa.Date	False False False
error_type	sa.String(128)	False False False
error_message	sa.Text	False False False
created_at	sa.DateTime(timezone=True)	False False False

39. historical_underlying_replay_outcome (16 columns)

Migration: m77_002_historical_underlying_replay_authority.py

Column	Type	Null / PK / Unique / FK-default metadata
prediction_id	sa.String(160)	None True False FK historical_underlying_replay_prediction.prediction_id
replay_run_id	sa.String(128)	False False False
symbol	sa.String(32)	False False False
as_of	sa.Date	False False False
status	sa.String(32)	False False False
entry_triggered	sa.Boolean	False False False
ambiguous_same_bar	sa.Boolean	False False False
mfe_pct	sa.Float	None False False
mae_pct	sa.Float	None False False
return_5d_pct	sa.Float	None False False
return_10d_pct	sa.Float	None False False
return_20d_pct	sa.Float	None False False
return_40d_pct	sa.Float	None False False
return_60d_pct	sa.Float	None False False
outcome_json	postgresql.JSONB	False False False
created_at	sa.DateTime(timezone=True)	False False False

Indexes: op.create_index('ix_m77_1_outcome_run', 'historical_underlying_replay_outcome', ['replay_run_id'])

40. historical_underlying_replay_prediction (12 columns)

Migration: m77_002_historical_underlying_replay_authority.py

Column	Type	Null / PK / Unique / FK-default metadata
prediction_id	sa.String(160)	None True False

Column	Type	Null / PK / Unique / FK-default metadata
replay_run_id	sa.String(128)	False False False FK historical_underlying_replay_run.replay_run_id
symbol	sa.String(32)	False False False
as_of	sa.Date	False False False
direction	sa.String(32)	False False False
primary_category	sa.String(64)	None False False
overall_score	sa.Float	False False False
confidence	sa.Float	False False False
state_hash	sa.String(64)	False False False
profile_json	postgresql.JSONB	False False False
lineage_json	postgresql.JSONB	False False False
created_at	sa.DateTime(timezone=True)	False False False

Indexes: op.create_index('ix_m77_1_pred_run', 'historical_underlying_replay_prediction', ['replay_run_id'])

41. historical_underlying_replay_run (13 columns)

Migration: m77_002_historical_underlying_replay_authority.py

Column	Type	Null / PK / Unique / FK-default metadata
replay_run_id	sa.String(128)	None True False
replay_mode	sa.String(64)	False False False
champion_mode	sa.String(96)	False False False
start_date	sa.Date	False False False
end_date	sa.Date	False False False
cadence	sa.String(16)	False False False
status	sa.String(32)	False False False
authority_version	sa.String(96)	False False False
started_at	sa.DateTime(timezone=True)	False False False
completed_at	sa.DateTime(timezone=True)	None False False
prediction_count	sa.Integer	False False False server_default=0
failure_count	sa.Integer	False False False server_default=0
metadata_json	postgresql.JSONB	False False False

42. institutional_decision_lineage (17 columns)

Migration: m47_001_persistent_lineage.py

Column	Type	Null / PK / Unique / FK-default metadata
decision_id	sa.String(96)	None True False
decision_run_id	sa.String(96)	False False False FK institutional_decision_lineage_run.decision_run_id
scanner_run_id	sa.String(96)	True False False
candidate_id	sa.String(96)	True False False
symbol	sa.String(32)	True False False
strategy	sa.String(96)	True False False
action	sa.String(48)	True False False
confidence	sa.Float()	False False False
accepted	sa.Boolean()	False False False
publication_name	sa.String(64)	True False False

Column	Type	Null / PK / Unique / FK-default metadata
ingestion_run_id	sa.String(96)	True False False
option_snapshot_id	sa.String(96)	True False False
market_state_hash	sa.String(64)	False False False
decision_engine_version	sa.String(48)	False False False
policy_version	sa.String(48)	False False False
payload_json	sa.Text()	False False False

Indexes: op.create_index('ix_decision_lineage_run', 'institutional_decision_lineage', ['decision_run_id'])

43. institutional_decision_lineage_run (16 columns)

Migration: m47_001_persistent_lineage.py

Column	Type	Null / PK / Unique / FK-default metadata
decision_run_id	sa.String(96)	None True False
publication_name	sa.String(64)	True False False
ingestion_run_id	sa.String(96)	True False False
publication_status	sa.String(24)	False False False
option_snapshot_timesta	sa.DateTime(timezone=True)	True False False
option_snapshot_id	sa.String(96)	True False False
published_state_degrade	sa.Boolean()	False False False server_default=sa.false()

Indexes: op.create_index('ix_decision_lineage_run_publication', 'institutional_decision_lineage_run', ['publication_name', 'started_at'])

44. institutional_event_forecast_snapshots (18 columns)

Migration: m69_005_event_outcomes_forecast_snapshots.py

Column	Type	Null / PK / Unique / FK-default metadata
forecast_snapshot_id	sa.String(180)	None True False
event_id	sa.String(128)	False False False
symbol	sa.String(32)	False False False
event_type	sa.String(64)	False False False
event_date	sa.Date()	False False False
event_session	sa.String(32)	None False False
snapshot_date	sa.Date()	False False False
snapshot_timestamp	sa.DateTime(timezone=True)	False False False
days_to_event	sa.Integer()	False False False
implied_move_pct	sa.Float()	None False False
historical_move_pct	sa.Float()	None False False
forecast_move_pct	sa.Float()	None False False
expected_move_pct	sa.Float()	None False False
confidence	sa.Float()	False False False
calculation_method	sa.String(96)	False False False
feature_hash	sa.String(64)	False False False
feature_json	postgresql.JSONB(astext_type=sa.Text)	False False
created_at	sa.DateTime(timezone=True)	False False False

Indexes: op.create_index('ix_m696_forecast_symbol_type_date', 'institutional_event_forecast_snapshots', ['symbol', 'event_type', 'event_date', 'snapshot_date'])

45. institutional_event_outcomes (25 columns)

Migration: m69_004_institutional_event_intelligence.py

Column	Type	Null / PK / Unique / FK-default metadata
outcome_id	sa.String(160)	None True False
event_id	sa.String(128)	False False False
observed_at	sa.String(64)	False False False
predicted_move_pct	sa.Float()	None False False
realized_move_pct	sa.Float()	None False False
forecast_error_pct	sa.Float()	None False False
payload_json	sa.JSON()	False False False
symbol	sa.String(32)	None False False
event_type	sa.String(64)	None False False
event_date	sa.Date()	None False False
event_session	sa.String(32)	None False False
status	sa.String(32)	False False False server_default=PROVISIONAL

Indexes: op.create_index('ix_m696_outcome_event', 'institutional_event_outcomes', ['event_id', 'observed_at'])

46. institutional_event_pricing_snapshots (10 columns)

Migration: m69_004_institutional_event_intelligence.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_id	sa.String(160)	None True False
event_id	sa.String(128)	False False False
snapshot_timestamp	sa.String(64)	False False False
implied_move_pct	sa.Float()	None False False
historical_move_pct	sa.Float()	None False False
forecast_move_pct	sa.Float()	None False False
expected_move_pct	sa.Float()	None False False
confidence	sa.Float()	False False False
classification	sa.String(48)	False False False
payload_json	sa.JSON()	False False False

Indexes: op.create_index('ix_m696_pricing_event', 'institutional_event_pricing_snapshots', ['event_id', 'snapshot_timestamp'])

47. institutional_inflection_publications (8 columns)

Migration: m68_001_institutional_inflection_intelligence.py | Model: src/trading_ai/inflection_intelligence/models.py

Column	Type	Null / PK / Unique / FK-default metadata
publication_id	sa.String(128)	None True False
publication_name	sa.String(128)	False False True
source_run_id	sa.String(128)	False False False
status	sa.String(32)	False False False
symbol_count	sa.Integer()	False False False
high_conviction_count	sa.Integer()	False False False
published_at	sa.String(64)	False False False
payload_json	sa.JSON()	False False False

Indexes: op.create_index(name, 'institutional_inflection_publications', [column])

48. institutional_inflection_snapshots (14 columns)

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_id	sa.String(128)	None True False
publication_name	sa.String(128)	False False False
source_run_id	sa.String(128)	False False False
symbol	sa.String(32)	False False False
timeframe	sa.String(16)	False False False
direction	sa.String(16)	False False False
transition_state	sa.String(64)	False False False
inflection_score	sa.Float()	False False False
confidence	sa.Float()	False False False
horizon_min_sessions	sa.Integer()	False False False
horizon_max_sessions	sa.Integer()	False False False
state_hash	sa.String(128)	False False False
snapshot_timestamp	sa.String(64)	False False False
payload_json	sa.JSON()	False False False

Indexes: op.create_index(name, 'institutional_inflection_snapshots', [col])

49. institutional_inflection_timeline_events (9 columns)

Migration: m68_001_institutional_inflection_intelligence.py | Model: src/trading_ai/inflection_intelligence/models.py

Column	Type	Null / PK / Unique / FK-default metadata
event_id	sa.String(128)	None True False
symbol	sa.String(32)	False False False
timeframe	sa.String(16)	False False False
transition_state	sa.String(64)	False False False
inflection_score	sa.Float()	False False False
confidence	sa.Float()	False False False
state_hash	sa.String(128)	False False False
event_timestamp	sa.String(64)	False False False
payload_json	sa.JSON()	False False False

Indexes: op.create_index('ix_m68_timeline_symbol', 'institutional_inflection_timeline_events', ['symbol'])

50. institutional_intelligence_snapshots (9 columns)

Migration: 20260730_m55_institutional_intelligence.py | Model: src/trading_ai/institutional_intelligence/models.py

Column	Type	Null / PK / Unique / FK-default metadata
intelligence_id	sa.String(128)	None True False
opportunity_id	sa.String(128)	False False False
opportunity_version	sa.Integer()	False False False
snapshot_id	sa.String(128)	False False False
snapshot_timestamp	sa.String(64)	False False False

Indexes: op.create_index('ix_institutional_intelligence_snapshots_{c}', 'institutional_intelligence_snapshots', [c])

51. institutional_market_context_evaluation (13 columns)

Migration: m46_002_polygon_authoritative_closure.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.BigInteger()	None True False
decision_id	sa.String(96)	False False False
symbol	sa.String(24)	False False False
snapshot_timestamp	sa.DateTime(timezone=True)	None False False
policy_version	sa.String(32)	False False False
outcome	sa.String(32)	False False False
allowed	sa.Boolean()	False False False
total_adjustment	sa.Float()	False False False
confidence	sa.Float()	False False False
reasons_json	sa.Text()	False False False
conflicts_json	sa.Text()	False False False
payload_json	sa.Text()	False False False
created_at	sa.DateTime(timezone=True)	False False False server_default=sa.func.now()

52. institutional_market_structure_snapshot (19 columns)

Migration: m44_001_institutional_market_structure.py

Column	Type	Null / PK / Unique / FK-default metadata
symbol	sa.String(16)	None True False
as_of_date	sa.Date()	None True False
option_snapshot_date	sa.Date()	False False False
spot	sa.Float()	False False False
net_gamma_exposure	sa.Float()	False False False
net_delta_exposure	sa.Float()	False False False
net_vanna_exposure	sa.Float()	False False False
net_charm_exposure	sa.Float()	False False False
gamma_regime	sa.String(32)	False False False
gamma_flip	sa.Float()	None False False
call_wall	sa.Float()	None False False
put_wall	sa.Float()	None False False
bull_probability	sa.Float()	False False False
bear_probability	sa.Float()	False False False
range_probability	sa.Float()	False False False
breakout_probability	sa.Float()	False False False
confidence	sa.String(16)	False False False
payload_json	sa.Text()	False False False

53. institutional_option_contract_recommendations (8 columns)

Migration: m62_001_institutional_options_domain.py | Model: src/trading_ai/institutional_options/models.py

Column	Type	Null / PK / Unique / FK-default metadata
contract_recommendatio	sa.String(128)	None True False
opportunity_id	sa.String(128)	False False False
strategy_candidate_id	sa.String(128)	False False False
option_snapshot_id	sa.String(128)	False False False
executable	sa.Boolean()	False False False server_default=sa.false()

Column	Type	Null / PK / Unique / FK-default metadata
liquidity_score	sa.Float()	True False False
created_at	sa.String(64)	False False False
payload_json	postgresql.JSONB(astext_type=sa.Text False	False False

Indexes: op.create_index('ix_m62_contract_opportunity', 'institutional_option_contract_recommendations', ['opportunity_id'])

54. institutional_option_decision_snapshots (16 columns)

Migration: m62_005_institutional_decision_snapshots.py | Model: src/trading_ai/institutional_options/models.py

Column	Type	Null / PK / Unique / FK-default metadata
decision_snapshot_id	sa.String(128)	None True False
opportunity_id	sa.String(128)	False False True
strategy_candidate_id	sa.String(128)	False False False
contract_recommendatio	sa.String(128)	False False False
valuation_id	sa.String(128)	False False False
management_snapshot_i	sa.String(128)	False False False
institutional_score	sa.Float()	False False False
calibrated_probability	sa.Float()	None False False
expected_value	sa.Float()	None False False
capital_required	sa.Float()	None False False
selected_strategy	sa.String(64)	False False False
policy_version	sa.String(64)	False False False
state_hash	sa.String(128)	False False True
created_at	sa.String(64)	False False False
payload_json	postgresql.JSONB(astext_type=sa.Text False	False False

Indexes: op.create_index('ix_m62_decision_{col}', 'institutional_option_decision_snapshots', [col])

55. institutional_option_edge_ledger (6 columns)

Migration: m69_001_option_valuation_mispricing.py | Model: src/trading_ai/option_valuation_intelligence/models.py

Column	Type	Null / PK / Unique / FK-default metadata
ledger_id	sa.String(128)	None True False
contract_recommendatio	sa.String(128)	False False False
opportunity_id	sa.String(128)	False False False
state_hash	sa.String(128)	False False False
observed_at	sa.String(64)	False False False
payload_json	sa.JSON()	False False False

Indexes: op.create_index('ix_m69_edge_contract', 'institutional_option_edge_ledger', ['contract_recommendation_id'])

56. institutional_option_execution_recommendations (9 columns)

Migration: m62_001_institutional_options_domain.py | Model: src/trading_ai/institutional_options/models.py

Column	Type	Null / PK / Unique / FK-default metadata
opportunity_id	sa.String(128)	False False True
strategy_candidate_id	sa.String(128)	False False False
contract_recommendatio	sa.String(128)	False False False
underlying_stop	sa.Float()	False False False
trailing_policy	sa.String(64)	False False False

Column	Type	Null / PK / Unique / FK-default metadata
ready_for_trade_builder	sa.Boolean()	False False False server_default=sa.false()
created_at	sa.String(64)	False False False
payload_json	postgresql.JSONB(astext_type=sa.Text)	False False

Indexes: op.create_index('ix_m62_execution_opportunity', 'institutional_option_execution_recommendations', ['opportunity_id'])

57. institutional_option_handoffs (14 columns)

Migration: m62_003_institutional_options_handoff.py | Model: src/trading_ai/institutional_options/models.py

Column	Type	Null / PK / Unique / FK-default metadata
handoff_id	sa.String(128)	None True False
opportunity_id	sa.String(128)	False False False
strategy_candidate_id	sa.String(128)	False False False
contract_recommendatio	sa.String(128)	False False False
account_id	sa.String(128)	False False False
trade_plan_id	sa.String(128)	None False False
execution_intent_id	sa.String(128)	None False False
status	sa.String(32)	False False False
overrides_json	sa.JSON()	False False False
lineage_json	sa.JSON()	False False False
payload_json	sa.JSON()	False False False
created_at	sa.String(64)	False False False
updated_at	sa.String(64)	False False False

Indexes: op.create_index('ix_m62_handoff_{column}', 'institutional_option_handoffs', [column])

58. institutional_option_learning_snapshots (10 columns)

Migration: m62_004_outcome_learning.py | Model: src/trading_ai/institutional_options/models.py

Column	Type	Null / PK / Unique / FK-default metadata
learning_snapshot_id	sa.String(128)	None True False
scope	sa.String(32)	False False False
scope_value	sa.String(128)	None False False
observation_count	sa.Integer()	False False False
win_rate	sa.Float()	None False False

Indexes: op.create_index('ix_m62_learning_{column}', 'institutional_option_learning_snapshots', [column])

59. institutional_option_management_snapshots (9 columns)

Migration: m62_002_strategy_valuation_management.py | Model: src/trading_ai/institutional_options/models.py

Column	Type	Null / PK / Unique / FK-default metadata
management_snapshot_i	sa.String(128)	None True False
opportunity_id	sa.String(128)	False False False
strategy_candidate_id	sa.String(128)	False False False
thesis_integrity	sa.Float()	False False False
position_health	sa.Float()	False False False
action	sa.String(32)	False False False
trailing_policy	sa.String(64)	False False False
created_at	sa.String(64)	False False False

Column	Type	Null / PK / Unique / FK-default metadata
payload_json	postgresql.JSONB(astext_type=sa.Text False	False False

Indexes: op.create_index('ix_m62_management_opportunity', 'institutional_option_management_snapshots', ['opportunity_id'])

60. institutional_option_opportunities (19 columns)

Migration: m62_001_institutional_options_domain.py | Model: src/trading_ai/institutional_options/models.py

Column	Type	Null / PK / Unique / FK-default metadata
opportunity_id	sa.String(128)	None True False
symbol	sa.String(32)	False False False
asset_class	sa.String(16)	False False False
state	sa.String(32)	False False False
direction	sa.String(16)	False False False
category	sa.String(64)	False False False
overall_score	sa.Float()	False False False
confidence	sa.Float()	False False False
conviction	sa.String(32)	False False False
thesis_id	sa.String(128)	False False False
stock_publication_name	sa.String(128)	False False False
stock_scanner_run_id	sa.String(128)	False False False
stock_candidate_id	sa.String(128)	False False False
stock_state_hash	sa.String(128)	False False False
option_snapshot_id	sa.String(128)	True False False
version	sa.Integer()	False False False server_default=1
created_at	sa.String(64)	False False False
updated_at	sa.String(64)	False False False
payload_json	postgresql.JSONB(astext_type=sa.Text False	False False

Indexes: op.create_index(name, 'institutional_option_opportunities', cols)

61. institutional_option_opportunity_audit (8 columns)

Migration: m62_001_institutional_options_domain.py | Model: src/trading_ai/institutional_options/models.py

Column	Type	Null / PK / Unique / FK-default metadata
audit_id	sa.String(128)	None True False
opportunity_id	sa.String(128)	False False False
previous_state	sa.String(32)	True False False
new_state	sa.String(32)	False False False
actor	sa.String(128)	False False False
reason	sa.Text()	False False False
event_timestamp	sa.String(64)	False False False
payload_json	postgresql.JSONB(astext_type=sa.Text False	False False

Indexes: op.create_index('ix_m62_audit_opportunity', 'institutional_option_opportunity_audit', ['opportunity_id'])

62. institutional_option_outcome_attributions (8 columns)

Migration: m62_001_institutional_options_domain.py | Model: src/trading_ai/institutional_options/models.py

Column	Type	Null / PK / Unique / FK-default metadata
attribution_id	sa.String(128)	None True False

Column	Type	Null / PK / Unique / FK-default metadata
opportunity_id	sa.String(128)	False False True
strategy_candidate_id	sa.String(128)	True False False
contract_recommendatio	sa.String(128)	True False False
outcome	sa.String(32)	True False False
realized_return_pct	sa.Float()	True False False
exit_reason	sa.String(64)	True False False
payload_json	postgresql.JSONB(astext_type=sa.Tex False	False False

Indexes: op.create_index('ix_m62_outcome_opportunity', 'institutional_option_outcome_attributions', ['opportunity_id'])

63. institutional_option_outcome_observations (14 columns)

Migration: m62_004_outcome_learning.py | Model: src/trading_ai/institutional_options/models.py

Column	Type	Null / PK / Unique / FK-default metadata
observation_id	sa.String(128)	None True False
opportunity_id	sa.String(128)	False False True
strategy_candidate_id	sa.String(128)	None False False
setup_category	sa.String(64)	False False False
market_regime	sa.String(64)	None False False

Indexes: op.create_index('ix_m62_outcome_observation_{column}', 'institutional_option_outcome_observations', [column])

64. institutional_option_relative_value_snapshots (13 columns)

Migration: m69_002_relative_value_event_mispricing.py | Model: src/trading_ai/option_valuation_intelligence/models.py

Column	Type	Null / PK / Unique / FK-default metadata
relative_value_id	sa.String(128)	None True False
valuation_run_id	sa.String(128)	False False False
contract_recommendatio	sa.String(128)	False False False
symbol	sa.String(32)	False False False
sector	sa.String(96)	False False False
peer_group	sa.String(160)	False False False
symbol_iv	sa.Float()	False False False
peer_median_iv	sa.Float()	False False False
divergence_pct	sa.Float()	False False False
z_score	sa.Float()	False False False
relationship_regime	sa.String(32)	False False False
snapshot_timestamp	sa.String(64)	False False False
payload_json	sa.JSON()	False False False

Indexes: op.create_index('ix_m69_rv_run', 'institutional_option_relative_value_snapshots', ['valuation_run_id'])

65. institutional_option_strategy_candidates (10 columns)

Migration: m62_001_institutional_options_domain.py | Model: src/trading_ai/institutional_options/models.py

Column	Type	Null / PK / Unique / FK-default metadata
strategy_candidate_id	sa.String(128)	None True False
opportunity_id	sa.String(128)	False False False
strategy	sa.String(64)	False False False
disposition	sa.String(32)	False False False

Column	Type	Null / PK / Unique / FK-default metadata
eligibility_score	sa.Float()	False False False
strategy_score	sa.Float()	True False False
complexity	sa.String(32)	False False False
rank	sa.Integer()	True False False
selected	sa.Boolean()	False False False server_default=sa.false()
payload_json	postgresql.JSONB(astext_type=sa.Text)	False False

Indexes: op.create_index('ix_m62_strategy_opportunity', 'institutional_option_strategy_candidates', ['opportunity_id'])

66. institutional_option_strategy_comparisons (6 columns)

Migration: m62_001_institutional_options_domain.py | Model: src/trading_ai/institutional_options/models.py

Column	Type	Null / PK / Unique / FK-default metadata
comparison_id	sa.String(128)	None True False
opportunity_id	sa.String(128)	False False True
policy_version	sa.String(64)	False False False
created_at	sa.String(64)	False False False
payload_json	postgresql.JSONB(astext_type=sa.Text)	False False

Indexes: op.create_index('ix_m62_comparison_opportunity', 'institutional_option_strategy_comparisons', ['opportunity_id'])

67. institutional_option_strategy_valuations (10 columns)

Migration: m62_002_strategy_valuation_management.py | Model: src/trading_ai/institutional_options/models.py

Column	Type	Null / PK / Unique / FK-default metadata
valuation_id	sa.String(128)	None True False
opportunity_id	sa.String(128)	False False False
strategy_candidate_id	sa.String(128)	False False False
strategy_score	sa.Float()	False False False
calibrated_probability	sa.Float()	True False False
expected_value	sa.Float()	True False False
expected_return_on_risk	sa.Float()	True False False
selected	sa.Boolean()	False False False server_default=sa.false()
created_at	sa.String(64)	False False False
payload_json	postgresql.JSONB(astext_type=sa.Text)	False False

Indexes: op.create_index('ix_m62_valuation_opportunity', 'institutional_option_strategy_valuations', ['opportunity_id'])

68. institutional_option_theses (10 columns)

Migration: m62_001_institutional_options_domain.py | Model: src/trading_ai/institutional_options/models.py

Column	Type	Null / PK / Unique / FK-default metadata
thesis_id	sa.String(128)	None True False
opportunity_id	sa.String(128)	False False True
direction	sa.String(16)	False False False
setup_category	sa.String(64)	False False False
primary_timeframe	sa.String(16)	False False False
invalidation_level	sa.Float()	False False False
entry_zone_low	sa.Float()	False False False

Indexes: op.create_index('ix_m62_theses_opportunity', 'institutional_option_theses', ['opportunity_id'])

69. institutional_option_valuation_events (11 columns)

Migration: m69_002_relative_value_event_mispricing.py | Model: src/trading_ai/option_valuation_intelligence/models.py

Column	Type	Null / PK / Unique / FK-default metadata
event_id	sa.String(128)	None True False
symbol	sa.String(32)	False False False
event_type	sa.String(48)	False False False
event_date	sa.String(32)	False False False
status	sa.String(24)	False False False
expected_move_pct	sa.Float()	None False False
historical_move_pct	sa.Float()	None False False
confidence	sa.Float()	False False False
source	sa.String(64)	False False False
payload_json	sa.JSON()	False False False
name	typ	True False False

Indexes: op.create_index('ix_m69_event_symbol_date', 'institutional_option_valuation_events', ['symbol', 'event_date'])

70. institutional_option_valuation_publications (8 columns)

Migration: m69_001_option_valuation_mispricing.py | Model: src/trading_ai/option_valuation_intelligence/models.py

Column	Type	Null / PK / Unique / FK-default metadata
publication_id	sa.String(128)	None True False
publication_name	sa.String(128)	False False True
status	sa.String(32)	False False False
contract_count	sa.Integer()	False False False
underpriced_count	sa.Integer()	False False False
overpriced_count	sa.Integer()	False False False
published_at	sa.String(64)	False False False
payload_json	sa.JSON()	False False False

71. institutional_option_valuation_snapshots (14 columns)

Migration: m69_001_option_valuation_mispricing.py | Model: src/trading_ai/option_valuation_intelligence/models.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_id	sa.String(128)	None True False
contract_recommendatio	sa.String(128)	False False False
opportunity_id	sa.String(128)	False False False

Indexes: op.create_index(n, 'institutional_option_valuation_snapshots', [c])

72. iv_surface_snapshot (13 columns)

Migration: m44_002_normalized_market_structure.py | Model: src/trading_ai/institutional_market_structure/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
symbol	sa.String(16)	None True False
as_of_date	sa.Date()	None True False
expiry	sa.Date()	None True False
strike	sa.Float()	None True False
option_type	sa.String(8)	None True False
dte	sa.Integer()	False False False

Column	Type	Null / PK / Unique / FK-default metadata
moneyness	sa.Float()	False False False
delta	sa.Float()	None False False
implied_volatility	sa.Float()	False False False
bid	sa.Float()	False False False
ask	sa.Float()	False False False
mid	sa.Float()	False False False
spread_pct	sa.Float()	None False False

73. live_trading_approvals (12 columns)

Migration: m67_001_live_trading_governance.py | Model: src/trading_ai/live_trading_governance/models.py

Column	Type	Null / PK / Unique / FK-default metadata
approval_id	sa.String(64)	None True False
portfolio_id	sa.String(64)	False False False
policy_id	sa.String(64)	False False False
approval_type	sa.String(32)	False False False
status	sa.String(32)	False False False
requested_by	sa.String(128)	False False False
approved_by	sa.String(128)	None False False
reason	sa.String(512)	False False False
expires_at	sa.DateTime(timezone=True)	None False False
payload_json	sa.JSON()	False False False
created_at	sa.DateTime(timezone=True)	False False False
updated_at	sa.DateTime(timezone=True)	False False False

Indexes: op.create_index('ix_m67_approval_portfolio', 'live_trading_approvals', ['portfolio_id'])

74. live_trading_audit_events (7 columns)

Migration: m67_001_live_trading_governance.py | Model: src/trading_ai/live_trading_governance/models.py

Column	Type	Null / PK / Unique / FK-default metadata
event_id	sa.String(64)	None True False
portfolio_id	sa.String(64)	False False False
event_type	sa.String(64)	False False False
actor	sa.String(128)	False False False
reason	sa.String(512)	False False False
payload_json	sa.JSON()	False False False
created_at	sa.DateTime(timezone=True)	False False False

Indexes: op.create_index('ix_m67_audit_portfolio', 'live_trading_audit_events', ['portfolio_id'])

75. live_trading_certification_runs (10 columns)

Migration: m67_001_live_trading_governance.py | Model: src/trading_ai/live_trading_governance/models.py

Column	Type	Null / PK / Unique / FK-default metadata
run_id	sa.String(64)	None True False
portfolio_id	sa.String(64)	False False False
policy_id	sa.String(64)	False False False
status	sa.String(32)	False False False

Column	Type	Null / PK / Unique / FK-default metadata
passed_checks	sa.Integer()	False False False
failed_checks	sa.Integer()	False False False
checks_json	sa.JSON()	False False False
evidence_json	sa.JSON()	False False False
started_at	sa.DateTime(timezone=True)	False False False
completed_at	sa.DateTime(timezone=True)	None False False

Indexes: op.create_index('ix_m67_cert_portfolio', 'live_trading_certification_runs', ['portfolio_id'])

76. live_trading_kill_switches (12 columns)

Migration: m67_001_live_trading_governance.py | Model: src/trading_ai/live_trading_governance/models.py

Column	Type	Null / PK / Unique / FK-default metadata
switch_id	sa.String(64)	None True False
portfolio_id	sa.String(64)	False False False
scope	sa.String(32)	False False False
scope_value	sa.String(128)	False False False
active	sa.Boolean()	False False False
action	sa.String(32)	False False False
reason	sa.String(512)	False False False
activated_by	sa.String(128)	False False False
activated_at	sa.DateTime(timezone=True)	False False False
cleared_by	sa.String(128)	None False False
cleared_at	sa.DateTime(timezone=True)	None False False
metadata_json	sa.JSON()	False False False

Indexes: op.create_index('ix_m67_kill_portfolio', 'live_trading_kill_switches', ['portfolio_id'])

77. live_trading_policies (17 columns)

Migration: m67_001_live_trading_governance.py | Model: src/trading_ai/live_trading_governance/models.py

Column	Type	Null / PK / Unique / FK-default metadata
policy_id	sa.String(64)	None True False
portfolio_id	sa.String(64)	False False False
version	sa.Integer()	False False False
status	sa.String(32)	False False False
environment	sa.String(16)	False False False
live_routing_enabled	sa.Boolean()	False False False
max_trade_loss_pct	sa.Float()	False False False
max_daily_loss_pct	sa.Float()	False False False
max_portfolio_heat_pct	sa.Float()	False False False
max_contracts	sa.Integer()	False False False
max_open_orders	sa.Integer()	False False False
allowed_symbols_json	sa.JSON()	False False False
allowed_strategies_json	sa.JSON()	False False False
metadata_json	sa.JSON()	False False False
created_at	sa.DateTime(timezone=True)	False False False
updated_at	sa.DateTime(timezone=True)	False False False

Indexes: op.create_index('ix_m67_policy_portfolio', 'live_trading_policies', ['portfolio_id'])

78. m73_exit_reservations (8 columns)

Migration: m73_001_autonomous_dynamic_position_management.py | Model: src/trading_ai/autonomous_position_management/models.py

Column	Type	Null / PK / Unique / FK-default metadata
reservation_id	sa.String(128)	None True False
position_id	sa.String(128)	False False False
instruction_id	sa.String(128)	False False False
status	sa.String(32)	False False False
reserved_quantity	sa.Integer()	False False False
created_at	sa.String(64)	False False False
updated_at	sa.String(64)	False False False
metadata_json	sa.JSON()	False False False

Indexes: op.create_index(n, 'm73_exit_reservations', [c])

79. m73_management_decisions (12 columns)

Migration: m73_001_autonomous_dynamic_position_management.py | Model: src/trading_ai/autonomous_position_management/models.py

Column	Type	Null / PK / Unique / FK-default metadata
decision_id	sa.String(128)	None True False
position_id	sa.String(128)	False False False
manager_id	sa.String(128)	False False False
cycle_timestamp	sa.String(64)	False False False
action	sa.String(40)	False False False
confidence	sa.Float()	False False False
conviction_score	sa.Float()	False False False
thesis_integrity	sa.Float()	False False False
current_stop	sa.Float()	None False False
current_target	sa.Float()	None False False
evidence_json	sa.JSON()	False False False
explanation	sa.Text()	False False False

Indexes: op.create_index(n, 'm73_management_decisions', [c])

80. m73_position_managers (13 columns)

Migration: m73_001_autonomous_dynamic_position_management.py | Model: src/trading_ai/autonomous_position_management/models.py

Column	Type	Null / PK / Unique / FK-default metadata
manager_id	sa.String(128)	None True False
position_id	sa.String(128)	False False True
portfolio_id	sa.String(128)	False False False
state	sa.String(40)	False False False
automation_mode	sa.String(40)	False False False
protection_state	sa.String(40)	False False False
heartbeat_at	sa.String(64)	False False False
activated_at	sa.String(64)	False False False
recovered_at	sa.String(64)	None False False
last_decision	sa.String(40)	None False False
conviction_score	sa.Float()	False False False

Column	Type	Null / PK / Unique / FK-default metadata
thesis_integrity	sa.Float()	False False False
metadata_json	sa.JSON()	False False False

Indexes: op.create_index(n, 'm73_position_managers', [c])

81. m73_replay_events (6 columns)

Migration: m73_001_autonomous_dynamic_position_management.py | Model: src/trading_ai/autonomous_position_management/models.py

Column	Type	Null / PK / Unique / FK-default metadata
event_id	sa.String(128)	None True False
position_id	sa.String(128)	False False False
event_timestamp	sa.String(64)	False False False
sequence_no	sa.Integer()	False False False
event_type	sa.String(64)	False False False
payload_json	sa.JSON()	False False False

Indexes: op.create_index(n, 'm73_replay_events', [c])

82. m77_13_cadence_states (11 columns)

Migration: m77_004_multi_cadence_certified_baseline_forward_shadow.py

Column	Type	Null / PK / Unique / FK-default metadata
state_id	sa.String(128)	None True False
cadence	sa.String(16)	False False False
as_of	sa.Date()	False False False
symbol	sa.String(32)	False False False
direction	sa.String(32)	False False False
overall_score	sa.Float()	None False False
confidence	sa.Float()	None False False
regime	sa.String(64)	False False False
state_hash	sa.String(128)	None False False
source_replay_run_id	sa.String(128)	False False False
payload_json	postgresql.JSONB()	False False False

Indexes: op.create_index('ix_m77_13_state_cadence_asof', 'm77_13_cadence_states', ['cadence', 'as_of'])

83. m77_13_forward_outcomes (9 columns)

Migration: m77_004_multi_cadence_certified_baseline_forward_shadow.py

Column	Type	Null / PK / Unique / FK-default metadata
outcome_id	sa.String(128)	None True False
signal_id	sa.String(128)	False False False FK m77_13_forward_signals.signal_id
target_session_date	sa.Date()	False False False
observed_at	sa.DateTime(timezone=True)	False False False
target_close	sa.Float()	False False False
raw_return_pct	sa.Float()	False False False
thesis_return_pct	sa.Float()	False False False
directional_hit	sa.Boolean()	False False False
payload_json	postgresql.JSONB()	False False False

Indexes: op.create_index('ix_m77_13_outcome_target_date', 'm77_13_forward_outcomes', ['target_session_date'])

84. m77_13_forward_signals (19 columns)

Migration: m77_004_multi_cadence_certified_baseline_forward_shadow.py

Column	Type	Null / PK / Unique / FK-default metadata
signal_id	sa.String(128)	None True False
signal_fingerprint	sa.String(64)	False False False
captured_at	sa.DateTime(timezone=True)	False False False
source_as_of	sa.Date()	False False False
symbol	sa.String(32)	False False False
baseline_source	sa.String(16)	False False False
baseline_id	sa.String(512)	False False False
horizon_sessions	sa.Integer()	False False False
direction	sa.String(32)	False False False
regime	sa.String(64)	False False False
score_band	sa.String(32)	False False False
confidence_band	sa.String(32)	False False False
reference_price	sa.Float()	False False False
daily_state_as_of	sa.Date()	None False False
weekly_state_as_of	sa.Date()	None False False
monthly_state_as_of	sa.Date()	None False False
policy_sha256	sa.String(64)	False False False
status	sa.String(32)	False False False
payload_json	postgresql.JSONB()	False False False

Indexes: op.create_index('ix_m77_13_signal_status', 'm77_13_forward_signals', ['status'])

85. m77_shadow_outcomes (9 columns)

Migration: m77_003_live_forward_shadow_intelligence.py

Column	Type	Null / PK / Unique / FK-default metadata
outcome_id	sa.String(128)	None True False
signal_id	sa.String(128)	False False False FK m77_shadow_signals.signal_id
target_session_date	sa.Date()	False False False
observed_at	sa.DateTime(timezone=True)	False False False
target_close	sa.Float()	False False False
raw_return_pct	sa.Float()	False False False
thesis_return_pct	sa.Float()	False False False
directional_hit	sa.Boolean()	False False False
payload_json	postgresql.JSONB()	False False False

Indexes: op.create_index('ix_m77_shadow_outcomes_target_session_date', 'm77_shadow_outcomes', ['target_session_date'])

86. m77_shadow_signals (21 columns)

Migration: m77_003_live_forward_shadow_intelligence.py

Column	Type	Null / PK / Unique / FK-default metadata
signal_id	sa.String(128)	None True False
signal_fingerprint	sa.String(64)	False False False
captured_at	sa.DateTime(timezone=True)	False False False
source_as_of_date	sa.Date()	False False False

Column	Type	Null / PK / Unique / FK-default metadata
scanner_run_id	sa.String(128)	False False False
candidate_id	sa.String(128)	None False False
symbol	sa.String(32)	False False False
direction	sa.String(32)	False False False
primary_category	sa.String(64)	False False False
structure	sa.String(64)	False False False
overall_score	sa.Float()	False False False
confidence	sa.Float()	None False False
historical_regime	sa.String(64)	False False False
shadow_tier	sa.String(32)	False False False
candidate_horizon_id	sa.String(512)	False False False
horizon_sessions	sa.Integer()	False False False
reference_price	sa.Float()	False False False
state_hash	sa.String(128)	None False False
policy_sha256	sa.String(64)	False False False
status	sa.String(32)	False False False
payload_json	postgresql.JSONB()	False False False

Indexes: `op.create_index('fix_m77_shadow_signals_{column}', 'm77_shadow_signals', [column])`

87. managed_position_events (8 columns)

Migration: 20260730_m57_portfolio_intelligence.py | Model: `src/trading_ai/portfolio_intelligence/models.py`

Column	Type	Null / PK / Unique / FK-default metadata
event_id	sa.String(128)	None True False
position_id	sa.String(128)	False False False
position_version	sa.Integer	False False False

88. managed_position_health_snapshots (8 columns)

Migration: 20260730_m57_portfolio_intelligence.py | Model: `src/trading_ai/portfolio_intelligence/models.py`

Column	Type	Null / PK / Unique / FK-default metadata
health_snapshot_id	sa.String(128)	None True False
position_id	sa.String(128)	False False False
position_version	sa.Integer	False False False
snapshot_timestamp	sa.String(64)	False False False
health_score	sa.Float	False False False
direction	sa.String(32)	False False False
confidence	sa.Float	False False False
payload_json	sa.JSON	False False False

89. managed_positions (22 columns)

Migration: 20260730_m57_portfolio_intelligence.py | Model: `src/trading_ai/portfolio_intelligence/models.py`

Column	Type	Null / PK / Unique / FK-default metadata
position_id	sa.String(128)	None True False
portfolio_id	sa.String(128)	False False False
trade_plan_id	sa.String(128)	False False False

Column	Type	Null / PK / Unique / FK-default metadata
opportunity_id	sa.String(128)	False False False
intelligence_id	sa.String(128)	None False False
execution_id	sa.String(128)	None False False
symbol	sa.String(32)	False False False
strategy	sa.String(64)	False False False
direction	sa.String(16)	False False False
state	sa.String(32)	False False False
version	sa.Integer	False False False
opened_at	sa.String(64)	False False False
closed_at	sa.String(64)	None False False
entry_value	sa.Float	False False False
realized_pnl	sa.Float	False False False
mark_json	sa.JSON	False False False
health_json	sa.JSON	False False False
decision_json	sa.JSON	False False False
metadata_json	sa.JSON	False False False
created_by	sa.String(128)	False False False
created_at	sa.String(64)	False False False
updated_at	sa.String(64)	False False False

90. market_breadth_snapshot (10 columns)

Migration: m45_001_market_overview.py | Model: src/trading_ai/market_overview/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_timestamp	sa.DateTime(timezone=True)	None True False
universe_name	sa.String(64)	None True False
as_of_date	sa.Date()	False False False

91. market_features (20 columns)

Migration: e1ebcb5044fb_sync_base_models.py, e8a1a4451024_add_market_features_table.py | Model: src/trading_ai/market/features.py

Column	Type	Null / PK / Unique / FK-default metadata
symbol	sa.String(16)	None True False
date	sa.Date	None True False
ema20	sa.Float	None False False
ema50	sa.Float	None False False
ema200	sa.Float	None False False
rsi14	sa.Float	None False False
macd	sa.Float	None False False
macd_signal	sa.Float	None False False
atr14	sa.Float	None False False
bollinger_width	sa.Float	None False False
hv20	sa.Float	None False False
volume_sma20	sa.Float	None False False
volume_ratio	sa.Float	None False False
vwap	sa.Float	None False False

Column	Type	Null / PK / Unique / FK-default metadata
price_vs_vwap	sa.Float	None False False
market_regime	sa.String(32)	None False False
Date	?	None True False
Float	?	None False False

92. market_ingestion_publication (13 columns)

Migration: m46_003_ingestion_publication.py

Column	Type	Null / PK / Unique / FK-default metadata
publication_name	sa.String(64)	None True False
run_id	sa.String(96)	False False False
published_at	sa.DateTime(timezone=True)	False False False
as_of_date	sa.Date()	False False False
market_intelligence_time	sa.DateTime(timezone=True)	False False False
option_snapshot_timesta	sa.DateTime(timezone=True)	True False False
option_snapshot_id	sa.String(96)	True False False
readiness_status	sa.String(24)	False False False
scanner_ready	sa.Boolean()	False False False server_default=sa.false()
decision_context_ready	sa.Boolean()	False False False server_default=sa.false()
details_json	sa.Text()	False False False
created_at	sa.DateTime(timezone=True)	False False False server_default=sa.func.now()
updated_at	sa.DateTime(timezone=True)	False False False server_default=sa.func.now()

Indexes: op.create_index('ix_market_ingestion_publication_published_at', 'market_ingestion_publication', ['published_at'])

93. market_intelligence_snapshot (9 columns)

Migration: m46_001_market_intelligence.py | Model: src/trading_ai/market_intelligence/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_timestamp	sa.DateTime(timezone=True)	None True False
as_of_date	sa.Date()	False False False
universe_name	sa.String(64)	False False False
source_snapshot_timesta	sa.DateTime(timezone=True)	False False False
confidence	sa.Float()	False False False
provenance	sa.String(32)	False False False
calculation_version	sa.String(32)	False False False
payload_json	sa.Text()	False False False
created_at	sa.DateTime(timezone=True)	False False False

94. market_internal_snapshot (0 columns)

Model: src/trading_ai/market_intelligence/database_models.py

Column detail was not safely parsed from the source-derived catalog; refer to the cited migration/model path above.

95. market_liquidity_snapshot (0 columns)

Model: src/trading_ai/market_intelligence/database_models.py

Column detail was not safely parsed from the source-derived catalog; refer to the cited migration/model path above.

96. market_opportunity_snapshot (11 columns)

Migration: m46_001_market_intelligence.py | Model: src/trading_ai/market_intelligence/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_timestamp	sa.DateTime(timezone=True)	None True False
opportunity_rank	sa.Integer()	None True False
as_of_date	sa.Date()	False False False
opportunity_type	sa.String(48)	False False False
instrument	sa.String(32)	False False False
direction	sa.String(16)	False False False
strategy_family	sa.String(48)	False False False
score	sa.Float()	False False False
confidence	sa.Float()	False False False
provenance	sa.String(32)	False False False
payload_json	sa.Text()	False False False

97. market_overview_snapshot (11 columns)

Migration: m45_001_market_overview.py | Model: src/trading_ai/market_overview/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_timestamp	sa.DateTime(timezone=True)	None True False
as_of_date	sa.Date()	False False False
market_bias	sa.String(32)	False False False
preferred_strategy	sa.String(64)	False False False
trend_regime	sa.String(32)	False False False
volatility_regime	sa.String(32)	False False False
breadth_regime	sa.String(32)	False False False
liquidity_regime	sa.String(32)	False False False
correlation_regime	sa.String(32)	False False False
regime_transition_risk	sa.String(16)	False False False
payload_json	sa.Text()	False False False

98. market_quote_event (12 columns)

Migration: m46_002_polygon_authoritative_closure.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.BigInteger()	None True False
provider	sa.String(32)	False False False
event_timestamp	sa.DateTime(timezone=True)	False False False
sequence_number	sa.BigInteger()	False False False
symbol	sa.String(64)	False False False
asset_type	sa.String(16)	False False False
bid_price	sa.Float()	None False False
bid_size	sa.Float()	None False False
ask_price	sa.Float()	None False False
ask_size	sa.Float()	None False False
bid_exchange	sa.String(16)	None False False
ask_exchange	sa.String(16)	None False False

Indexes: op.create_index('ix_market_quote_event_symbol_ts', 'market_quote_event', ['symbol', 'event_timestamp'])

99. market_risk_snapshot (0 columns)

Model: src/trading_ai/market_intelligence/database_models.py

Column detail was not safely parsed from the source-derived catalog; refer to the cited migration/model path above.

100. market_sentiment_snapshot (9 columns)

Migration: m46_001_market_intelligence.py | Model: src/trading_ai/market_intelligence/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_timestamp	sa.DateTime(timezone=True)	None True False
as_of_date	sa.Date()	False False False
overall_sentiment_score	sa.Float()	False False False
risk_appetite_score	sa.Float()	False False False
fear_score	sa.Float()	False False False
sentiment_label	sa.String(24)	False False False
confidence	sa.Float()	False False False
provenance	sa.String(32)	False False False
payload_json	sa.Text()	False False False

101. market_trade_event (10 columns)

Migration: m46_002_polygon_authoritative_closure.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.BigInteger()	None True False
provider	sa.String(32)	False False False
event_timestamp	sa.DateTime(timezone=True)	False False False
sequence_number	sa.BigInteger()	False False False
symbol	sa.String(64)	False False False
asset_type	sa.String(16)	False False False
price	sa.Float()	False False False
size	sa.Float()	False False False
exchange	sa.String(16)	None False False
conditions_json	sa.Text()	False False False server_default=[]

Indexes: op.create_index('ix_market_trade_event_symbol_ts', 'market_trade_event', ['symbol', 'event_timestamp'])

102. market_volatility_snapshot (0 columns)

Model: src/trading_ai/market_intelligence/database_models.py

Column detail was not safely parsed from the source-derived catalog; refer to the cited migration/model path above.

103. microstructure_liquidity_snapshot (12 columns)

Migration: m46_002_polygon_authoritative_closure.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_timestamp	sa.DateTime(timezone=True)	None True False
symbol	sa.String(64)	None True False
as_of_date	sa.Date()	False False False
executable_quote_pct	sa.Float()	None False False
median_relative_spread_	sa.Float()	None False False
average_trade_size	sa.Float()	None False False
liquidity_score	sa.Float()	False False False

Column	Type	Null / PK / Unique / FK-default metadata
liquidity_regime	sa.String(24)	False False False
depth_available	sa.Boolean()	False False False server_default=sa.false()
confidence	sa.Float()	False False False
provenance	sa.String(48)	False False False
payload_json	sa.Text()	False False False

104. name (7 columns)

Migration: m46_001_market_intelligence.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_timestamp	sa.DateTime(timezone=True)	None True False
as_of_date	sa.Date()	False False False
score	sa.Float()	False False False
regime	sa.String(32)	False False False
confidence	sa.Float()	False False False
provenance	sa.String(32)	False False False
payload_json	sa.Text()	False False False

105. opex_forecast_outcomes (11 columns)

Migration: m71_001_opex_intelligence.py | Model: src/trading_ai/opex_intelligence/models.py

Column	Type	Null / PK / Unique / FK-default metadata
outcome_id	sa.String(128)	None True False
forecast_id	sa.String(128)	False False True
symbol	sa.String(16)	False False False
expiration	sa.String(16)	False False False
settlement_price	sa.Float()	False False False
in_50	sa.Integer()	False False False
in_68	sa.Integer()	False False False
in_90	sa.Integer()	False False False
magnet_distance_pct	sa.Float()	None False False
realized_at	sa.String(64)	False False False
payload_json	sa.JSON()	False False False

Indexes: op.create_index(name, 'opex_forecast_outcomes', [column])

106. opex_forecast_publications (8 columns)

Migration: m71_001_opex_intelligence.py | Model: src/trading_ai/opex_intelligence/models.py

Column	Type	Null / PK / Unique / FK-default metadata
publication_id	sa.String(128)	None True False
publication_name	sa.String(96)	False False True
status	sa.String(24)	False False False
published_at	sa.String(64)	False False False
forecast_count	sa.Integer()	False False False
payload_json	sa.JSON()	False False False
coverage_status	sa.String(32)	False False False server_default=UNKNOWN

Indexes: op.create_index('ix_m714_opex_publication_authority_fingerprint', 'opex_forecast_publications', ['authority_input_fingerprint'])

107. opex_forecast_snapshots (28 columns)

Migration: m71_001_opex_intelligence.py | Model: src/trading_ai/opex_intelligence/models.py

Column	Type	Null / PK / Unique / FK-default metadata
forecast_id	sa.String(128)	None True False
symbol	sa.String(16)	False False False
expiration	sa.String(16)	False False False
cycle_type	sa.String(24)	False False False
dte	sa.Integer()	False False False
forecast_timestamp	sa.String(64)	False False False
source_as_of_date	sa.String(16)	False False False
spot	sa.Float()	False False False
range50_low	sa.Float()	False False False
range50_high	sa.Float()	False False False
range68_low	sa.Float()	False False False
range68_high	sa.Float()	False False False
range90_low	sa.Float()	False False False
range90_high	sa.Float()	False False False
magnet	sa.Float()	None False False
magnet_probability	sa.Float()	False False False
support	sa.Float()	None False False
resistance	sa.Float()	None False False
gamma_flip_current	sa.Float()	None False False
gamma_flip_forecast	sa.Float()	None False False
call_wall_current	sa.Float()	None False False
call_wall_forecast	sa.Float()	None False False
put_wall_current	sa.Float()	None False False
put_wall_forecast	sa.Float()	None False False
dealer_pressure	sa.Float()	False False False
confidence	sa.Float()	False False False
payload_json	sa.JSON()	False False False
input_fingerprint	sa.String(64)	True False False

Indexes: op.create_index('ix_m714_opex_forecast_input_fingerprint', 'opex_forecast_snapshots', ['input_fingerprint'])

108. opex_settlement_values (9 columns)

Migration: m71_004_governed_opex_authority.py | Model: src/trading_ai/opex_intelligence/models.py

Column	Type	Null / PK / Unique / FK-default metadata
settlement_id	sa.String(128)	None True False
underlying_symbol	sa.String(16)	False False False
expiration	sa.String(16)	False False False
settlement_symbol	sa.String(32)	False False False
settlement_style	sa.String(64)	False False False
settlement_value	sa.Float()	False False False
settlement_source	sa.String(96)	False False False
observed_at	sa.String(64)	False False False
lineage_json	sa.JSON()	False False False

Indexes: op.create_index(name, 'opex_settlement_values', [column])

109. opportunities (14 columns)

Migration: m54_001_canonical_opportunity_domain.py | Model: src/trading_ai/opportunity_domain/models.py

Column	Type	Null / PK / Unique / FK-default metadata
opportunity_id	sa.String(128)	None True False
scanner_run_id	sa.String(128)	False False False
snapshot_id	sa.String(128)	False False False
snapshot_timestamp	sa.String(64)	False False False
symbol	sa.String(32)	False False False
direction	sa.String(16)	False False False
strategy	sa.String(64)	False False False
workflow_state	sa.String(32)	False False False
version	sa.Integer()	False False False
source_payload_json	sa.JSON()	False False False
created_by	sa.String(128)	False False False
created_at	sa.String(64)	False False False
updated_at	sa.String(64)	False False False
metadata_json	sa.JSON()	False False False

Indexes: op.create_index('ix_opportunities_{col}', 'opportunities', [col])

110. opportunity_audit_events (10 columns)

Migration: m54_001_canonical_opportunity_domain.py | Model: src/trading_ai/opportunity_domain/models.py

Column	Type	Null / PK / Unique / FK-default metadata
event_id	sa.String(128)	None True False
opportunity_id	sa.String(128)	False False False
opportunity_version	sa.Integer()	False False False
event_type	sa.String(64)	False False False
previous_state	sa.String(32)	None False False
new_state	sa.String(32)	False False False
actor	sa.String(128)	False False False
reason	sa.Text()	False False False

Indexes: op.create_index('ix_opportunity_audit_events_{col}', 'opportunity_audit_events', [col])

111. option_contract_history (21 columns)

Migration: 1b0a35cdc5fb_add_option_contract_history_table.py | Model: src/trading_ai/market/option_models.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.Integer()	None True False
underlying_symbol	sa.String()	False False False
option_symbol	sa.String()	False False False
quote_date	sa.Date()	False False False
expiry	sa.Date()	False False False
option_type	sa.String()	False False False
strike	sa.Float()	False False False
bid	sa.Float()	False False False server_default=0
ask	sa.Float()	False False False server_default=0

Column	Type	Null / PK / Unique / FK-default metadata
mid	sa.Float()	False False False server_default=0
last	sa.Float()	False False False server_default=0
volume	sa.Integer()	False False False server_default=0
open_interest	sa.Integer()	False False False server_default=0
implied_volatility	sa.Float()	False False False server_default=0
delta	sa.Float()	False False False server_default=0
gamma	sa.Float()	False False False server_default=0
theta	sa.Float()	False False False server_default=0
vega	sa.Float()	False False False server_default=0
rho	sa.Float()	False False False server_default=0
quote_timestamp	sa.String(length=64)	True False False
source_underlying_price	sa.Float()	True False False

Indexes: op.create_index('ix_option_chain_lookup', 'option_contract_history', ['underlying_symbol', 'quote_date', 'option_type', 'expiry', 'strike'])

112. option_contract_snapshot (27 columns)

Migration: m46_002_polygon_authoritative_closure.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.BigInteger()	None True False
snapshot_run_id	sa.BigInteger()	False False False FK option_snapshot_run.id
snapshot_timestamp	sa.DateTime(timezone=True)	False False False
underlying_symbol	sa.String(24)	False False False
option_symbol	sa.String(64)	False False False
expiry	sa.Date()	False False False
option_type	sa.String(8)	False False False
strike	sa.Float()	False False False
bid	sa.Float()	None False False
ask	sa.Float()	None False False
last	sa.Float()	None False False
mark	sa.Float()	None False False
volume	sa.BigInteger()	None False False

Indexes: op.create_index('ix_option_contract_snapshot_lookup', 'option_contract_snapshot', ['underlying_symbol', 'snapshot_timestamp'])

113. option_snapshot_run (15 columns)

Migration: m46_002_polygon_authoritative_closure.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.BigInteger()	None True False
snapshot_id	sa.String(96)	False False False
snapshot_timestamp	sa.DateTime(timezone=True)	False False False
as_of_date	sa.Date()	False False False
provider	sa.String(32)	False False False server_default=POLYGON
capture_status	sa.String(24)	False False False
is_partial	sa.Boolean()	False False False server_default=sa.false()
completeness_score	sa.Float()	False False False server_default=0
symbols_requested	sa.Integer()	False False False server_default=0

Column	Type	Null / PK / Unique / FK-default metadata
symbols_succeeded	sa.Integer()	False False False server_default=0
symbols_failed	sa.Integer()	False False False server_default=0
contracts_received	sa.Integer()	False False False server_default=0
contracts_persisted	sa.Integer()	False False False server_default=0
warnings_json	sa.Text()	False False False server_default=[]
created_at	sa.DateTime(timezone=True)	False False False server_default=sa.func.now()

Indexes: op.create_index('ix_option_snapshot_run_timestamp', 'option_snapshot_run', ['snapshot_timestamp'])

114. outcome_probability_audit_events (7 columns)

Migration: m77_001_governed_outcome_probability.py | Model: src/trading_ai/outcome_probability/models.py

Column	Type	Null / PK / Unique / FK-default metadata
event_id	sa.String(160)	None True False
entity_id	sa.String(160)	False False False
event_type	sa.String(64)	False False False
actor	sa.String(128)	False False False
reason	sa.Text()	False False False
occurred_at	sa.String(64)	False False False
payload_json	sa.JSON()	False False False

Indexes: op.create_index('ix_m77_audit_entity', 'outcome_probability_audit_events', ['entity_id'])

115. outcome_probability_model_artifacts (17 columns)

Migration: m77_001_governed_outcome_probability.py | Model: src/trading_ai/outcome_probability/models.py

Column	Type	Null / PK / Unique / FK-default metadata
model_id	sa.String(160)	None True False
model_version	sa.String(96)	False False True
lifecycle_state	sa.String(32)	False False False
feature_version	sa.String(96)	False False False
label_version	sa.String(96)	False False False
training_started_at	sa.String(64)	False False False
training_completed_at	sa.String(64)	False False False
training_cutoff	sa.String(64)	False False False
sample_size	sa.Integer()	False False False
artifact_json	sa.JSON()	False False False
evaluation_json	sa.JSON()	False False False
governance_json	sa.JSON()	False False False
state_hash	sa.String(64)	False False False
approved_by	sa.String(128)	None False False
approved_at	sa.String(64)	None False False
activated_by	sa.String(128)	None False False
activated_at	sa.String(64)	None False False

Indexes: op.create_index('uq_m77_one_shadow_active', 'outcome_probability_model_artifacts', ['lifecycle_state'], unique=True, postgresql_where=sa.text("lifecycle_state =

116. outcome_probability_observations (24 columns)

Migration: m77_001_governed_outcome_probability.py | Model: src/trading_ai/outcome_probability/models.py

Column	Type	Null / PK / Unique / FK-default metadata
observation_id	sa.String(160)	None True False
candidate_id	sa.String(128)	False False False
scanner_run_id	sa.String(128)	False False False
symbol	sa.String(32)	False False False
as_of	sa.String(64)	False False False
horizon_end	sa.String(64)	None False False
status	sa.String(48)	False False False
label_version	sa.String(96)	False False False
feature_version	sa.String(96)	False False False
entry_triggered	sa.Integer()	None False False
target_1_before_stop	sa.Integer()	None False False
target_2_before_stop	sa.Integer()	None False False
target_3_before_stop	sa.Integer()	None False False
profitable_at_horizon	sa.Integer()	None False False
thesis_invalidation	sa.Integer()	None False False
maximum_favorable_exc	sa.Float()	None False False
realized_return_pct	sa.Float()	None False False
days_to_target_1	sa.Integer()	None False False

Indexes: op.create_index(name, 'outcome_probability_observations', [column])

117. outcome_probability_predictions (13 columns)

Migration: m77_001_governed_outcome_probability.py | Model: src/trading_ai/outcome_probability/models.py

Column	Type	Null / PK / Unique / FK-default metadata
prediction_id	sa.String(160)	None True False
candidate_id	sa.String(128)	False False False
scanner_run_id	sa.String(128)	False False False
symbol	sa.String(32)	False False False
model_id	sa.String(160)	False False False
predicted_at	sa.String(64)	False False False
mode	sa.String(24)	False False False
recommended_dispositio	sa.String(24)	False False False
target_1_probability	sa.Float()	None False False
profitable_probability	sa.Float()	None False False
uncertainty	sa.Float()	False False False
assessment_json	sa.JSON()	False False False
state_hash	sa.String(64)	False False False

Indexes: op.create_index(name, 'outcome_probability_predictions', [column])

118. paper_automation_checkpoints (8 columns)

Migration: m49_001_authoritative_paper_trading.py | Model: src/trading_ai/authoritative_paper_trading/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
checkpoint_id	sa.String(128)	None True False
session_id	sa.String(128)	False False False
cycle_id	sa.String(128)	False False False

Column	Type	Null / PK / Unique / FK-default metadata
stage	sa.String(64)	False False False
status	sa.String(32)	False False False
payload_json	sa.JSON()	False False False
created_at	sa.String(64)	False False False
updated_at	sa.String(64)	False False False

Indexes: op.create_index('ix_paper_automation_checkpoints_{col}', 'paper_automation_checkpoints', [col])

119. paper_executions (22 columns)

Migration: m49_001_authoritative_paper_trading.py | Model: src/trading_ai/authoritative_paper_trading/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
execution_key	sa.String(160)	None True False
session_id	sa.String(128)	False False False
cycle_id	sa.String(128)	False False False
aggregate_id	sa.String(128)	False False False
client_order_id	sa.String(128)	False False False
account_id	sa.String(64)	False False False
order_type	sa.String(32)	False False False
time_in_force	sa.String(32)	False False False
status	sa.String(32)	False False False
requested_quantity	sa.Float()	False False False
filled_quantity	sa.Float()	False False False
remaining_quantity	sa.Float()	False False False
average_fill_price	sa.Float()	None False False
gross_value	sa.Float()	False False False
commissions	sa.Float()	False False False
net_cash_flow	sa.Float()	False False False
latency_ms	sa.Integer()	False False False
rejection_reasons_json	sa.JSON()	False False False
warnings_json	sa.JSON()	False False False
created_at	sa.String(64)	False False False
updated_at	sa.String(64)	False False False
metadata_json	sa.JSON()	False False False

Indexes: op.create_index('ix_paper_executions_{col}', 'paper_executions', [col])

120. paper_fills (16 columns)

Migration: m49_001_authoritative_paper_trading.py | Model: src/trading_ai/authoritative_paper_trading/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
fill_id	sa.String(128)	None True False
execution_key	sa.String(160)	False False False
aggregate_id	sa.String(128)	False False False
client_order_id	sa.String(128)	False False False
leg_id	sa.String(128)	False False False
symbol	sa.String(64)	False False False
side	sa.String(16)	False False False

Column	Type	Null / PK / Unique / FK-default metadata
quantity	sa.Float()	False False False
fill_price	sa.Float()	False False False
reference_price	sa.Float()	False False False
slippage_amount	sa.Float()	False False False
slippage_bps	sa.Float()	False False False
commission	sa.Float()	False False False
latency_ms	sa.Integer()	False False False
filled_at	sa.String(64)	False False False
metadata_json	sa.JSON()	False False False

Indexes: `op.create_index('fix_paper_fills_{col}', 'paper_fills', [col])`

121. paper_position_lifecycle_events (10 columns)

Migration: `m49_001_authoritative_paper_trading.py` | Model: `src/trading_ai/authoritative_paper_trading/database_models.py`

Column	Type	Null / PK / Unique / FK-default metadata
event_id	sa.String(128)	None True False
position_id	sa.String(128)	False False False
portfolio_id	sa.String(64)	False False False
event_type	sa.String(64)	False False False
quantity	sa.Float()	False False False
price	sa.Float()	False False False
realized_pnl	sa.Float()	False False False server_default=0
occurred_at	sa.String(64)	False False False
reference_id	sa.String(128)	False False False
metadata_json	sa.JSON()	False False False

Indexes: `op.create_index('fix_paper_position_lifecycle_events_{col}', 'paper_position_lifecycle_events', [col])`

122. paper_position_marks (9 columns)

Migration: `m49_001_authoritative_paper_trading.py` | Model: `src/trading_ai/authoritative_paper_trading/database_models.py`

Column	Type	Null / PK / Unique / FK-default metadata
mark_id	sa.String(128)	None True False
position_id	sa.String(128)	False False False
portfolio_id	sa.String(64)	False False False
mark_price	sa.Float()	False False False
market_value	sa.Float()	False False False
unrealized_pnl	sa.Float()	False False False
marked_at	sa.String(64)	False False False
source	sa.String(64)	False False False
metadata_json	sa.JSON()	False False False

Indexes: `op.create_index('fix_paper_position_marks_{col}', 'paper_position_marks', [col])`

123. paper_trading_controls (8 columns)

Migration: `m49_001_authoritative_paper_trading.py` | Model: `src/trading_ai/authoritative_paper_trading/database_models.py`

Column	Type	Null / PK / Unique / FK-default metadata
account_id	sa.String(64)	None True False

Column	Type	Null / PK / Unique / FK-default metadata
entries_enabled	sa.Boolean()	False False False server_default=sa.true()
exits_enabled	sa.Boolean()	False False False server_default=sa.true()
trading_halted	sa.Boolean()	False False False server_default=sa.false()
halt_reason	sa.Text()	None False False
version	sa.Integer()	False False False server_default=1
updated_at	sa.String(64)	False False False
metadata_json	sa.JSON()	False False False

124. paper_trading_sessions (9 columns)

Migration: m49_001_authoritative_paper_trading.py | Model: src/trading_ai/authoritative_paper_trading/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
session_id	sa.String(128)	None True False
account_id	sa.String(64)	False False False
state	sa.String(32)	False False False
version	sa.Integer()	False False False server_default=1
started_at	sa.String(64)	False False False
updated_at	sa.String(64)	False False False
stopped_at	sa.String(64)	None False False
profile_json	sa.JSON()	False False False

Indexes: op.create_index('ix_paper_trading_sessions_account_id', 'paper_trading_sessions', ['account_id'])

125. performance_attribution_snapshots (5 columns)

Migration: m65_001_performance_command_center.py | Model: src/trading_ai/performance_learning/models.py

Column	Type	Null / PK / Unique / FK-default metadata
attribution_snapshot_id	sa.String(128)	None True False
portfolio_id	sa.String(128)	False False False
generated_at	sa.String(64)	False False False
sample_size	sa.Integer()	False False False
payload_json	sa.JSON()	False False False

126. performance_calibration_runs (12 columns)

Migration: m72_001_outcome_calibration_execution_quality.py | Model: src/trading_ai/performance_learning/models.py

Column	Type	Null / PK / Unique / FK-default metadata
calibration_run_id	sa.String(160)	None True False
scope	sa.String(64)	False False False
scope_value	sa.String(128)	False False False
target_type	sa.String(64)	False False False
sample_size	sa.Integer()	False False False
brier_score	sa.Float()	None False False
log_loss	sa.Float()	None False False
calibration_slope	sa.Float()	None False False
calibration_intercept	sa.Float()	None False False
generated_at	sa.String(64)	False False False
metrics_json	sa.JSON()	False False False

Indexes: op.create_index(n, 'performance_calibration_runs', [c])

127. performance_calibration_snapshots (10 columns)

Migration: m65_001_performance_command_center.py | Model: src/trading_ai/performance_learning/models.py

Column	Type	Null / PK / Unique / FK-default metadata
calibration_snapshot_id	sa.String(128)	None True False
portfolio_id	sa.String(128)	False False False
scope	sa.String(64)	False False False
scope_value	sa.String(128)	False False False
sample_size	sa.Integer()	False False False
brier_score	sa.Float()	False False False
log_loss	sa.Float()	False False False
generated_at	sa.String(64)	False False False
payload_json	sa.JSON()	False False False

128. performance_counterfactual_outcomes (11 columns)

Migration: m65_001_performance_command_center.py | Model: src/trading_ai/performance_learning/models.py

Column	Type	Null / PK / Unique / FK-default metadata
counterfactual_id	sa.String(128)	None True False
portfolio_id	sa.String(128)	False False False
opportunity_id	sa.String(128)	False False False
decision_state_hash	sa.String(128)	False False False
decision	sa.String(32)	False False False
selected	sa.Integer()	False False False
predicted_probability	sa.Float()	False False False
observed_return_pct	sa.Float()	None False False
evaluation_status	sa.String(32)	False False False
evaluated_at	sa.String(64)	False False False
payload_json	sa.JSON()	False False False

129. performance_execution_quality_snapshots (9 columns)

Migration: m72_001_outcome_calibration_execution_quality.py | Model: src/trading_ai/performance_learning/models.py

Column	Type	Null / PK / Unique / FK-default metadata
portfolio_id	sa.String(128)	False False False
generated_at	sa.String(64)	False False False
sample_size	sa.Integer()	False False False
average_quality_score	sa.Float()	False False False
average_slippage_pct	sa.Float()	False False False
edge_preservation_pct	sa.Float()	False False False
fill_rate_pct	sa.Float()	False False False
metrics_json	sa.JSON()	False False False

Indexes: op.create_index('ix_m72_excecq_portfolio', 'performance_execution_quality_snapshots', ['portfolio_id'])

130. performance_learning_audit_events (7 columns)

Migration: 20260730_m58_performance_learning.py | Model: src/trading_ai/performance_learning/models.py

Column	Type	Null / PK / Unique / FK-default metadata
event_id	sa.String(128)	None True False
entity_id	sa.String(128)	False False False
event_type	sa.String(64)	False False False
actor	sa.String(128)	False False False
reason	sa.Text	False False False
event_timestamp	sa.String(64)	False False False
payload_json	sa.JSON	False False False

131. performance_learning_observations (14 columns)

Migration: 20260730_m58_performance_learning.py | Model: src/trading_ai/performance_learning/models.py

Column	Type	Null / PK / Unique / FK-default metadata
observation_id	sa.String(128)	None True False
position_id	sa.String(128)	False False False
position_version	sa.Integer	False False False
portfolio_id	sa.String(128)	False False False
opportunity_id	sa.String(128)	False False False

132. performance_learning_policies (13 columns)

Migration: 20260730_m58_performance_learning.py | Model: src/trading_ai/performance_learning/models.py

Column	Type	Null / PK / Unique / FK-default metadata
policy_id	sa.String(128)	None True False
policy_name	sa.String(128)	False False False
version	sa.Integer	False False False
state	sa.String(32)	False False False
parameters_json	sa.JSON	False False False
evidence_json	sa.JSON	False False False
reason	sa.Text	False False False
created_by	sa.String(128)	False False False
approved_by	sa.String(128)	None False False
created_at	sa.String(64)	False False False
updated_at	sa.String(64)	False False False
activated_at	sa.String(64)	None False False

133. performance_learning_publications (8 columns)

Migration: m65_001_performance_command_center.py | Model: src/trading_ai/performance_learning/models.py

Column	Type	Null / PK / Unique / FK-default metadata
publication_id	sa.String(128)	None True False
publication_name	sa.String(128)	False False False
portfolio_id	sa.String(128)	False False False
report_id	sa.String(128)	False False False
status	sa.String(32)	False False False
sample_size	sa.Integer()	False False False
published_at	sa.String(64)	False False False
payload_json	sa.JSON()	False False False

134. performance_learning_reports (8 columns)

Migration: 20260730_m58_performance_learning.py | Model: src/trading_ai/performance_learning/models.py

Column	Type	Null / PK / Unique / FK-default metadata
report_id	sa.String(128)	None True False
portfolio_id	sa.String(128)	False False False
window_start	sa.String(64)	None False False
window_end	sa.String(64)	None False False

135. performance_prediction_outcomes (8 columns)

Migration: m72_001_outcome_calibration_execution_quality.py | Model: src/trading_ai/performance_learning/models.py

Column	Type	Null / PK / Unique / FK-default metadata
outcome_id	sa.String(160)	None True False
prediction_id	sa.String(160)	False False False
outcome_type	sa.String(64)	False False False
binary_outcome	sa.Integer()	None False False
realized_value	sa.Float()	None False False
error_value	sa.Float()	None False False
realized_at	sa.String(64)	False False False
outcome_json	sa.JSON()	False False False

Indexes: op.create_index(n, 'performance_prediction_outcomes', [c])

136. performance_prediction_registry (15 columns)

Migration: m72_001_outcome_calibration_execution_quality.py | Model: src/trading_ai/performance_learning/models.py

Column	Type	Null / PK / Unique / FK-default metadata
prediction_id	sa.String(160)	None True False
source_type	sa.String(48)	False False False
source_id	sa.String(160)	False False True
portfolio_id	sa.String(128)	None False False
symbol	sa.String(32)	False False False
strategy	sa.String(64)	None False False
target_type	sa.String(64)	False False False
predicted_probability	sa.Float()	None False False
confidence	sa.Float()	None False False
model_version	sa.String(96)	False False False
generated_at	sa.String(64)	False False False
horizon_end	sa.String(64)	None False False
features_json	sa.JSON()	False False False
prediction_json	sa.JSON()	False False False
lineage_json	sa.JSON()	False False False

Indexes: op.create_index(n, 'performance_prediction_registry', [c])

137. performance_trade_outcomes (20 columns)

Migration: m65_001_performance_command_center.py | Model: src/trading_ai/performance_learning/models.py

Column	Type	Null / PK / Unique / FK-default metadata
outcome_id	sa.String(128)	None True False

Column	Type	Null / PK / Unique / FK-default metadata
position_id	sa.String(128)	False False False
position_version	sa.Integer()	False False False
portfolio_id	sa.String(128)	False False False

138. portfolio_accounts (7 columns)

Migration: f36a1s5_portfolio_management_tables.py | Model: src/trading_ai/portfolio_management/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
portfolio_id	sa.String(64)	None True False
name	sa.String(255)	False False False
base_currency	sa.String(8)	False False False
initial_capital	sa.Float()	False False False
status	sa.String(32)	False False False
created_at	sa.String(64)	False False False
metadata_json	sa.JSON()	False False False

139. portfolio_action_recommendations (10 columns)

Migration: m64_003_cumulative_portfolio_optimization.py | Model: src/trading_ai/portfolio_risk_allocation/models.py

Column	Type	Null / PK / Unique / FK-default metadata
recommendation_id	sa.String(128)	None True False
portfolio_id	sa.String(64)	False False False
risk_snapshot_id	sa.String(128)	False False False
action_key	sa.String(255)	False False False
action_type	sa.String(32)	False False False
symbol	sa.String(32)	None False False
priority	sa.String(24)	False False False
status	sa.String(32)	False False False
created_at	sa.String(64)	False False False
payload_json	sa.JSON()	False False False

Indexes: op.create_index('ix_m64_action_portfolio', 'portfolio_action_recommendations', ['portfolio_id'])

140. portfolio_allocation_publications (8 columns)

Migration: m64_003_cumulative_portfolio_optimization.py | Model: src/trading_ai/portfolio_risk_allocation/models.py

Column	Type	Null / PK / Unique / FK-default metadata
publication_id	sa.String(128)	None True False
publication_name	sa.String(128)	False False False
portfolio_id	sa.String(64)	False False False
risk_snapshot_id	sa.String(128)	False False False
optimization_snapshot_id	sa.String(128)	False False False
published_at	sa.String(64)	False False False
status	sa.String(32)	False False False
payload_json	sa.JSON()	False False False

Indexes: op.create_index('ix_m64_publication_portfolio', 'portfolio_allocation_publications', ['portfolio_id'])

141. portfolio_audit_history (15 columns)

Migration: f36a1s5_portfolio_management_tables.py | Model: src/trading_ai/portfolio_management/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
audit_id	sa.String(128)	None True False
portfolio_id	sa.String(64)	False False False
snapshot_id	sa.String(128)	False False False
event_type	sa.String(64)	False False False
occurred_at	sa.String(64)	False False False
registry_fingerprint	sa.String(128)	False False False
open_position_count	sa.Integer()	False False False
closed_position_count	sa.Integer()	False False False
cash_balance	sa.Float()	False False False
net_liquidation_value	sa.Float()	False False False
capital_committed	sa.Float()	False False False
realized_pnl	sa.Float()	False False False
unrealized_pnl	sa.Float()	False False False
source_registry	sa.Text()	False False False
metadata_json	sa.JSON()	False False False

142. portfolio_cash_ledger (8 columns)

Migration: f36a1s5_portfolio_management_tables.py | Model: src/trading_ai/portfolio_management/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
entry_id	sa.String(128)	None True False
portfolio_id	sa.String(64)	False False False
event_type	sa.String(64)	False False False
amount	sa.Float()	False False False
balance_after	sa.Float()	False False False
occurred_at	sa.String(64)	False False False
reference_id	sa.String(128)	False False False
notes	sa.Text()	False False False

143. portfolio_cash_reservations (9 columns)

Migration: m49_001_authoritative_paper_trading.py | Model: src/trading_ai/authoritative_paper_trading/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
reservation_id	sa.String(128)	None True False
portfolio_id	sa.String(64)	False False False

Indexes: op.create_index(fix_portfolio_cash_reservations_{col}', 'portfolio_cash_reservations', [col])

144. portfolio_correlation_snapshots (5 columns)

Migration: m64_002_decision_intelligence.py | Model: src/trading_ai/portfolio_risk_allocation/models.py

Column	Type	Null / PK / Unique / FK-default metadata
correlation_snapshot_id	sa.String(128)	None True False
portfolio_id	sa.String(64)	False False False
risk_snapshot_id	sa.String(128)	False False False
generated_at	sa.String(64)	False False False
payload_json	sa.JSON()	False False False

Indexes: op.create_index('ix_m64_corr_portfolio', 'portfolio_correlation_snapshots', ['portfolio_id'])

145. portfolio_decision_intelligence_snapshots (15 columns)

Migration: m64_002_decision_intelligence.py | Model: src/trading_ai/portfolio_risk_allocation/models.py

Column	Type	Null / PK / Unique / FK-default metadata
decision_intelligence_id	sa.String(128)	None True False
portfolio_id	sa.String(64)	False False False
opportunity_id	sa.String(128)	False False False
risk_snapshot_id	sa.String(128)	False False False
portfolio_fit_score	sa.Float()	False False False
opportunity_cost_score	sa.Float()	False False False
final_portfolio_score	sa.Float()	False False False
recommended_quantity	sa.Integer()	False False False
recommended_capital	sa.Float()	False False False
decision	sa.String(32)	False False False
rank	sa.Integer()	None False False
state_hash	sa.String(128)	False False False
created_at	sa.String(64)	False False False
payload_json	sa.JSON()	False False False

Indexes: op.create_index(name, 'portfolio_decision_intelligence_snapshots', [col])

146. portfolio_fit_assessments (11 columns)

Migration: m64_001_portfolio_risk_allocation.py | Model: src/trading_ai/portfolio_risk_allocation/models.py

Column	Type	Null / PK / Unique / FK-default metadata
assessment_id	sa.String(128)	None True False
portfolio_id	sa.String(64)	False False False
candidate_id	sa.String(160)	False False False
risk_snapshot_id	sa.String(128)	False False False
symbol	sa.String(32)	False False False
portfolio_fit_score	sa.Float()	False False False
recommended_quantity	sa.Integer()	False False False
recommended_capital	sa.Float()	False False False
decision	sa.String(32)	False False False
assessed_at	sa.String(64)	False False False
payload_json	sa.JSON()	False False False

147. portfolio_intelligence_snapshots (5 columns)

Migration: 20260730_m57_portfolio_intelligence.py | Model: src/trading_ai/portfolio_intelligence/models.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_id	sa.String(128)	None True False
portfolio_id	sa.String(128)	False False False
snapshot_timestamp	sa.String(64)	False False False
payload_json	sa.JSON	False False False
generated_by	sa.String(128)	False False False

148. portfolio_optimization_snapshots (10 columns)

Migration: m64_003_cumulative_portfolio_optimization.py | Model: src/trading_ai/portfolio_risk_allocation/models.py

Column	Type	Null / PK / Unique / FK-default metadata
optimization_snapshot_id	sa.String(128)	None True False
portfolio_id	sa.String(64)	False False False
risk_snapshot_id	sa.String(128)	False False False
generated_at	sa.String(64)	False False False
status	sa.String(32)	False False False
objective_score	sa.Float()	False False False
selected_count	sa.Integer()	False False False
recommended_capital	sa.Float()	False False False
state_hash	sa.String(128)	False False False
payload_json	sa.JSON()	False False False

Indexes: op.create_index('ix_m64_optimizer_portfolio', 'portfolio_optimization_snapshots', ['portfolio_id'])

149. portfolio_positions (28 columns)

Migration: f36a1s5_portfolio_management_tables.py | Model: src/trading_ai/portfolio_management/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
position_id	sa.String(128)	None True False
portfolio_id	sa.String(64)	False False False
symbol	sa.String(32)	False False False

150. portfolio_risk_allocation_snapshots (16 columns)

Migration: m64_001_portfolio_risk_allocation.py | Model: src/trading_ai/portfolio_risk_allocation/models.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_id	sa.String(128)	None True False
portfolio_id	sa.String(64)	False False False
snapshot_timestamp	sa.String(64)	False False False
broker_publication_id	sa.String(128)	None False False
status	sa.String(32)	False False False
health_score	sa.Float()	False False False
net_liquidation	sa.Float()	False False False
buying_power	sa.Float()	False False False
capital_committed	sa.Float()	False False False
open_risk	sa.Float()	False False False
var_95	sa.Float()	False False False
expected_shortfall_95	sa.Float()	False False False
portfolio_heat_pct	sa.Float()	False False False
concentration_score	sa.Float()	False False False
diversification_score	sa.Float()	False False False
payload_json	sa.JSON()	False False False

Indexes: op.create_index('ix_m64_risk_portfolio', 'portfolio_risk_allocation_snapshots', ['portfolio_id', 'snapshot_timestamp'])

151. portfolio_risk_assessments (10 columns)

Migration: m37risk_portfolio_risk_management.py | Model: src/trading_ai/portfolio_risk_management/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
assessment_id	sa.String(length=128)	None True False
portfolio_id	sa.String(length=64)	False False False
generated_at	sa.String(length=64)	False False False
status	sa.String(length=32)	False False False
trading_control	sa.String(length=64)	False False False
net_liquidation_value	sa.Float()	False False False
cash_balance	sa.Float()	False False False
capital_committed	sa.Float()	False False False
open_position_count	sa.Integer()	False False False
payload_json	sa.JSON()	False False False

Indexes: op.create_index('ix_portfolio_risk_assessments_portfolio_id', 'portfolio_risk_assessments', ['portfolio_id'])

152. portfolio_risk_breaches (13 columns)

Migration: m37risk_portfolio_risk_management.py | Model: src/trading_ai/portfolio_risk_management/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
breach_id	sa.String(length=128)	None True False
assessment_id	sa.String(length=128)	False False False
portfolio_id	sa.String(length=64)	False False False
code	sa.String(length=128)	False False False
severity	sa.String(length=32)	False False False
status	sa.String(length=32)	False False False
observed_value	sa.Float()	False False False
limit_value	sa.Float()	False False False
recommended_action	sa.String(length=128)	False False False
message	sa.Text()	False False False
created_at	sa.String(length=64)	False False False
resolved_at	sa.String(length=64)	True False False
resolution	sa.Text()	True False False

Indexes: op.create_index('ix_portfolio_risk_breaches_assessment_id', 'portfolio_risk_breaches', ['assessment_id'])

153. portfolio_risk_budget_snapshots (7 columns)

Migration: m64_003_cumulative_portfolio_optimization.py | Model: src/trading_ai/portfolio_risk_allocation/models.py

Column	Type	Null / PK / Unique / FK-default metadata
budget_snapshot_id	sa.String(128)	None True False
portfolio_id	sa.String(64)	False False False
risk_snapshot_id	sa.String(128)	False False False
generated_at	sa.String(64)	False False False
status	sa.String(32)	False False False
utilization_pct	sa.Float()	False False False
payload_json	sa.JSON()	False False False

Indexes: op.create_index('ix_m64_budget_portfolio', 'portfolio_risk_budget_snapshots', ['portfolio_id'])

154. portfolio_snapshots (7 columns)

Migration: f36a1s5_portfolio_management_tables.py | Model: src/trading_ai/portfolio_management/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_id	sa.String(128)	None True False
portfolio_id	sa.String(64)	False False False
generated_at	sa.String(64)	False False False
registry_fingerprint	sa.String(128)	False False False
registry_json	sa.JSON()	False False False
exposure_json	sa.JSON()	False False False
warnings_json	sa.JSON()	False False False

155. portfolio_stress_allocation_snapshots (7 columns)

Migration: m64_001_portfolio_risk_allocation.py | Model: src/trading_ai/portfolio_risk_allocation/models.py

Column	Type	Null / PK / Unique / FK-default metadata
stress_snapshot_id	sa.String(128)	None True False
portfolio_id	sa.String(64)	False False False
risk_snapshot_id	sa.String(128)	False False False
generated_at	sa.String(64)	False False False
worst_scenario	sa.String(64)	False False False
worst_loss	sa.Float()	False False False
payload_json	sa.JSON()	False False False

156. portfolio_stress_results (8 columns)

Migration: m37risk_portfolio_risk_management.py | Model: src/trading_ai/portfolio_risk_management/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
result_id	sa.String(length=192)	None True False
assessment_id	sa.String(length=128)	False False False
portfolio_id	sa.String(length=64)	False False False
scenario_id	sa.String(length=128)	False False False
estimated_pnl	sa.Float()	False False False
estimated_loss_pct_nav	sa.Float()	False False False
status	sa.String(length=32)	False False False
payload_json	sa.JSON()	False False False

Indexes: op.create_index('ix_portfolio_stress_results_assessment_id', 'portfolio_stress_results', ['assessment_id'])

157. portfolio_sync_runs (12 columns)

Migration: f36a1s5_portfolio_management_tables.py | Model: src/trading_ai/portfolio_management/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
sync_id	sa.String(128)	None True False
portfolio_id	sa.String(64)	False False False
started_at	sa.String(64)	False False False
completed_at	sa.String(64)	False False False
status	sa.String(32)	False False False
registry_fingerprint	sa.String(128)	False False False

158. position_attribution_reports (5 columns)

Migration: 20260730_m57_portfolio_intelligence.py | Model: src/trading_ai/portfolio_intelligence/models.py

Column	Type	Null / PK / Unique / FK-default metadata
attribution_id	sa.String(128)	None True False
position_id	sa.String(128)	False False False
generated_at	sa.String(64)	False False False
payload_json	sa.JSON	False False False
generated_by	sa.String(128)	False False False

159. position_exit_instructions (9 columns)

Migration: m39pos_position_monitoring.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.Integer()	None True False
instruction_id	sa.String(64)	False False True
assessment_id	sa.String(64)	False False False
position_id	sa.String(128)	False False False
action	sa.String(32)	False False False
quantity	sa.Integer()	False False False
status	sa.String(32)	False False False
payload	sa.JSON()	False False False
created_at	sa.String(64)	False False False

Indexes: op.create_index('ix_m39_instruction_position', 'position_exit_instructions', ['position_id'])

160. position_monitoring_assessments (10 columns)

Migration: m39pos_position_monitoring.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.Integer()	None True False
assessment_id	sa.String(64)	False False True
position_id	sa.String(128)	False False False
portfolio_id	sa.String(128)	False False False
symbol	sa.String(32)	False False False
decision	sa.String(32)	False False False
urgency	sa.String(32)	False False False
return_pct	sa.Float()	False False False
payload	sa.JSON()	False False False
generated_at	sa.String(64)	False False False

Indexes: op.create_index('ix_m39_assessment_position', 'position_monitoring_assessments', ['position_id'])

161. price_history (3 columns)

Model: src/trading_ai/database/models.py

Column	Type	Null / PK / Unique / FK-default metadata
Column	Type	Null PK Unique Foreign key / default
Date	?	None True False
Float	?	None False False

162. production_api_audit_events (10 columns)

Migration: m40api_production_api_audit.py | Model: src/trading_ai/production_api/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.Integer()	None True False
event_id	sa.String(length=64)	False False True
request_id	sa.String(length=64)	False False False
method	sa.String(length=16)	False False False
path	sa.String(length=512)	False False False
status_code	sa.Integer()	False False False
duration_ms	sa.Float()	False False False
actor	sa.String(length=128)	True False False
occurred_at	sa.DateTime(timezone=True)	False False False
event_metadata	sa.JSON()	False False False

Indexes: op.create_index('ix_m40_api_request_id', 'production_api_audit_events', ['request_id'])

163. production_ops_alerts (14 columns)

Migration: m66_001_production_operations.py | Model: src/trading_ai/production_operations/models.py

Column	Type	Null / PK / Unique / FK-default metadata
alert_id	sa.String(128)	None True False
fingerprint	sa.String(160)	False False False
severity	sa.String(24)	False False False
status	sa.String(24)	False False False
category	sa.String(64)	False False False
title	sa.String(256)	False False False
message	sa.Text()	False False False
owner	sa.String(128)	False False False
recommended_action	sa.Text()	False False False
created_at	sa.String(64)	False False False
updated_at	sa.String(64)	False False False
acknowledged_at	sa.String(64)	None False False
acknowledged_by	sa.String(128)	None False False
payload_json	sa.JSON()	False False False

164. production_ops_audit_events (7 columns)

Migration: m66_001_production_operations.py | Model: src/trading_ai/production_operations/models.py

Column	Type	Null / PK / Unique / FK-default metadata
event_id	sa.String(128)	None True False
entity_type	sa.String(64)	False False False
entity_id	sa.String(160)	False False False
event_type	sa.String(96)	False False False
actor	sa.String(128)	False False False
occurred_at	sa.String(64)	False False False
payload_json	sa.JSON()	False False False

165. production_ops_locks (6 columns)

Migration: m66_001_production_operations.py | Model: src/trading_ai/production_operations/models.py

Column	Type	Null / PK / Unique / FK-default metadata
lock_name	sa.String(160)	None True False
owner_id	sa.String(160)	False False False
acquired_at	sa.String(64)	False False False
heartbeat_at	sa.String(64)	False False False
expires_at	sa.String(64)	False False False
metadata_json	sa.JSON()	False False False

166. production_ops_publication_freshness (10 columns)

Migration: m66_001_production_operations.py | Model: src/trading_ai/production_operations/models.py

Column	Type	Null / PK / Unique / FK-default metadata
freshness_id	sa.String(128)	None True False
publication_name	sa.String(128)	False False False
source_id	sa.String(160)	False False False
status	sa.String(32)	False False False
published_at	sa.String(64)	None False False
age_seconds	sa.Float()	None False False
maximum_age_seconds	sa.Integer()	False False False
reason	sa.Text()	False False False
lineage_json	sa.JSON()	False False False
evaluated_at	sa.String(64)	False False False

167. production_ops_readiness_publications (12 columns)

Migration: m66_001_production_operations.py | Model: src/trading_ai/production_operations/models.py

Column	Type	Null / PK / Unique / FK-default metadata
publication_id	sa.String(128)	None True False
publication_name	sa.String(128)	False False False
status	sa.String(32)	False False False
scanner_ready	sa.Boolean()	False False False
decision_ready	sa.Boolean()	False False False
execution_ready	sa.Boolean()	False False False
management_ready	sa.Boolean()	False False False
portfolio_ready	sa.Boolean()	False False False
learning_ready	sa.Boolean()	False False False
platform_ready	sa.Boolean()	False False False
published_at	sa.String(64)	False False False
payload_json	sa.JSON()	False False False

168. production_ops_recovery_actions (10 columns)

Migration: m66_001_production_operations.py | Model: src/trading_ai/production_operations/models.py

Column	Type	Null / PK / Unique / FK-default metadata
recovery_id	sa.String(128)	None True False
action_type	sa.String(64)	False False False
target_type	sa.String(64)	False False False
target_id	sa.String(160)	False False False

Column	Type	Null / PK / Unique / FK-default metadata
status	sa.String(32)	False False False
actor	sa.String(128)	False False False
reason	sa.Text()	False False False

169. production_ops_services (14 columns)

Migration: m66_001_production_operations.py | Model: src/trading_ai/production_operations/models.py

Column	Type	Null / PK / Unique / FK-default metadata
service_id	sa.String(96)	None True False
name	sa.String(160)	False False False
domain	sa.String(64)	False False False
status	sa.String(32)	False False False
critical	sa.Boolean()	False False False
heartbeat_at	sa.String(64)	None False False
last_success_at	sa.String(64)	None False False
last_failure_at	sa.String(64)	None False False
next_run_at	sa.String(64)	None False False
latency_ms	sa.Float()	None False False
retry_count	sa.Integer()	False False False
dependencies_json	sa.JSON()	False False False
metadata_json	sa.JSON()	False False False
updated_at	sa.String(64)	False False False

170. production_ops_workflow_runs (11 columns)

Migration: m66_001_production_operations.py | Model: src/trading_ai/production_operations/models.py

Column	Type	Null / PK / Unique / FK-default metadata
run_id	sa.String(128)	None True False
workflow_name	sa.String(128)	False False False
mode	sa.String(32)	False False False
status	sa.String(32)	False False False
started_at	sa.String(64)	False False False
finished_at	sa.String(64)	None False False
actor	sa.String(128)	False False False
current_stage	sa.String(128)	None False False
stage_results_json	sa.JSON()	False False False
error	sa.Text()	None False False
metadata_json	sa.JSON()	False False False

171. realtime_alert_events (7 columns)

Migration: m42ops_realtime_monitoring.py

Column	Type	Null / PK / Unique / FK-default metadata
alert_id	sa.String(128)	None True False
rule_id	sa.String(128)	False False False
severity	sa.String(32)	False False False
status	sa.String(32)	False False False

Column	Type	Null / PK / Unique / FK-default metadata
source	sa.String(64)	False False False
payload_json	sa.JSON()	False False False
created_at	sa.DateTime(timezone=True)	False False False

172. scanner_candidate_lineage (15 columns)

Migration: m47_001_persistent_lineage.py

Column	Type	Null / PK / Unique / FK-default metadata
candidate_id	sa.String(96)	None True False
scanner_run_id	sa.String(96)	False False False FK scanner_lineage_run.scanner_run_id

Indexes: op.create_index('ix_scanner_candidate_lineage_run_rank', 'scanner_candidate_lineage', ['scanner_run_id', 'rank'])

173. scanner_lineage_run (18 columns)

Migration: m47_001_persistent_lineage.py

Column	Type	Null / PK / Unique / FK-default metadata
scanner_run_id	sa.String(96)	None True False
publication_name	sa.String(64)	True False False
ingestion_run_id	sa.String(96)	True False False
publication_status	sa.String(24)	False False False
published_at	sa.DateTime(timezone=True)	True False False
market_as_of_date	sa.Date()	True False False
option_snapshot_timesta	sa.DateTime(timezone=True)	True False False
option_snapshot_id	sa.String(96)	True False False
published_state_degrade	sa.Boolean()	False False False server_default=sa.false()
scanner_version	sa.String(48)	False False False
market_state_hash	sa.String(64)	False False False
started_at	sa.DateTime(timezone=True)	False False False
completed_at	sa.DateTime(timezone=True)	False False False
status	sa.String(24)	False False False
candidate_count	sa.Integer()	False False False
metadata_json	sa.Text()	False False False

Indexes: op.create_index('ix_scanner_lineage_run_publication', 'scanner_lineage_run', ['publication_name', 'published_at'])

174. sector_breadth_snapshot (11 columns)

Migration: m46_001_market_intelligence.py | Model: src/trading_ai/market_intelligence/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_timestamp	sa.DateTime(timezone=True)	None True False
sector	sa.String(96)	None True False
as_of_date	sa.Date()	False False False
sector_ett	sa.String(16)	None False False
constituent_count	sa.Integer()	False False False
breadth_score	sa.Float()	False False False

175. sector_membership (13 columns)

Migration: m46_001_market_intelligence.py | Model: src/trading_ai/market_intelligence/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
symbol	sa.String(16)	None True False
effective_from	sa.Date()	None True False
company_name	sa.String(255)	False False False
sector	sa.String(96)	False False False
industry	sa.String(128)	None False False
sub_industry	sa.String(128)	None False False
sector_etf	sa.String(16)	None False False
classification_source	sa.String(255)	False False False
effective_to	sa.Date()	None False False
is_active	sa.Boolean()	False False False
confidence	sa.Float()	False False False
last_verified_at	sa.DateTime(timezone=True)	False False False
payload_json	sa.Text()	False False False

176. sector_rotation_snapshot (7 columns)

Migration: m45_001_market_overview.py | Model: src/trading_ai/market_overview/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_timestamp	sa.DateTime(timezone=True)	None True False
sector_etf	sa.String(16)	None True False
as_of_date	sa.Date()	False False False
sector	sa.String(64)	False False False
dealer_positioning_score	sa.Float()	None False False
rotation_label	sa.String(24)	False False False
payload_json	sa.Text()	False False False

177. sentiment_component_snapshot (10 columns)

Migration: m46_001_market_intelligence.py | Model: src/trading_ai/market_intelligence/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_timestamp	sa.DateTime(timezone=True)	None True False
component_name	sa.String(64)	None True False
as_of_date	sa.Date()	False False False
score	sa.Float()	False False False
weight	sa.Float()	False False False
contribution	sa.Float()	False False False
direction	sa.String(16)	False False False
confidence	sa.Float()	False False False
provenance	sa.String(32)	False False False
payload_json	sa.Text()	False False False

178. stock_accumulation_distribution_snapshots (8 columns)

Migration: m61_003_stock_intelligence.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.String(128)	None True False

179. stock_breakout_snapshots (8 columns)

Migration: m61_004_stock_intelligence.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.String(128)	None True False
symbol	sa.String(32)	False False False
scanner_run_id	sa.String(128)	False False False
candidate_id	sa.String(128)	True False False
snapshot_timestamp	sa.String(64)	False False False
payload_json	sa.JSON()	False False False
state	sa.String(32)	False False False
confirmation	sa.Float	False False False

180. stock_context_snapshots (8 columns)

Migration: m61_005_stock_intelligence.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.String(128)	None True False
symbol	sa.String(32)	False False False
scanner_run_id	sa.String(128)	False False False
candidate_id	sa.String(128)	True False False
snapshot_timestamp	sa.String(64)	False False False
payload_json	sa.JSON()	False False False
context_score	sa.Float	False False False
context_adjustment	sa.Float	False False False

181. stock_institutional_structure_zones (15 columns)

Migration: m61_010_institutional_structure_zones.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.String(128)	None True False
symbol	sa.String(32)	False False False
scanner_run_id	sa.String(128)	False False False
candidate_id	sa.String(128)	True False False
snapshot_timestamp	sa.String(64)	False False False
zone_type	sa.String(32)	False False False
lower_bound	sa.Float()	False False False
upper_bound	sa.Float()	False False False
representative_price	sa.Float()	False False False
strength	sa.Float()	False False False
confluence_score	sa.Float()	False False False
primary_timeframe	sa.String(16)	False False False
contributing_timeframes	postgresql.JSONB(astext_type=sa.Text)	False False
components	postgresql.JSONB(astext_type=sa.Text)	False False
payload_json	postgresql.JSONB(astext_type=sa.Text)	False False

Indexes: op.create_index('ix_stock_institutional_structure_zones_symbol', 'stock_institutional_structure_zones', ['symbol'])

182. stock_institutional_trend_snapshot (10 columns)

Migration: m52_004_institutional_trend_participation.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.BigInteger()	None True False
symbol	sa.String(length=32)	False False False
as_of_date	sa.Date()	False False False
snapshot_timestamp	sa.DateTime(timezone=True)	False False False
status	sa.String(length=32)	False False False
participation_score	sa.Float()	False False False
leadership_score	sa.Float()	False False False
trend_quality_score	sa.Float()	False False False
deterioration_risk_score	sa.Float()	False False False
payload_json	postgresql.JSONB(astext_type=sa.Text)	False False

Indexes: op.create_index('ix_stock_inst_trend_symbol_timestamp', 'stock_institutional_trend_snapshot', ['symbol', 'snapshot_timestamp'], unique=False)

183. stock_management_policy_performance (4 columns)

Migration: m61_008_outcome_learning.py

Column	Type	Null / PK / Unique / FK-default metadata
policy_name	sa.String(128)	False False False
observation_count	sa.Integer	False False False server_default=0
expectancy_pct	sa.Float	False False False server_default=0
score	sa.Float	False False False server_default=0

184. stock_opportunity_score_snapshots (9 columns)

Migration: m61_006_stock_intelligence.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.String(128)	None True False
symbol	sa.String(32)	False False False
scanner_run_id	sa.String(128)	False False False
candidate_id	sa.String(128)	True False False
snapshot_timestamp	sa.String(64)	False False False
payload_json	sa.JSON()	False False False
primary_category	sa.String(64)	False False False
overall_score	sa.Float	False False False
confidence	sa.Float	False False False

185. stock_outcome_attribution_snapshots (3 columns)

Migration: m61_008_outcome_learning.py

Column	Type	Null / PK / Unique / FK-default metadata
attribution_type	sa.String(64)	False False False
attribution_key	sa.String(128)	False False False
observation_count	sa.Integer	False False False server_default=0

186. stock_outcome_observations (7 columns)

Migration: m61_008_outcome_learning.py

Column	Type	Null / PK / Unique / FK-default metadata
outcome	sa.String(32)	False False False

187. stock_position_intelligence_snapshots (9 columns)

Migration: m61_007_position_intelligence.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.String(128)	None True False
symbol	sa.String(32)	False False False
scanner_run_id	sa.String(128)	False False False
candidate_id	sa.String(128)	True False False
snapshot_timestamp	sa.String(64)	False False False
payload_json	sa.JSON()	False False False
action	sa.String(32)	False False False
thesis_integrity	sa.Float	False False False
management_quality	sa.Float	False False False

188. stock_probability_calibration_snapshots (5 columns)

Migration: m61_008_outcome_learning.py

Column	Type	Null / PK / Unique / FK-default metadata
model_family	sa.String(128)	False False False
model_version	sa.String(128)	False False False
observation_count	sa.Integer	False False False server_default=0
brier_score	sa.Float	True False False

189. stock_scanner_candidates (8 columns)

Migration: m61_001_stock_intelligence.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.String(128)	None True False
symbol	sa.String(32)	False False False
scanner_run_id	sa.String(128)	False False False
candidate_id	sa.String(128)	True False False
snapshot_timestamp	sa.String(64)	False False False
payload_json	sa.JSON()	False False False
category	sa.String(64)	False False False
score	sa.Float	False False False

190. stock_scanner_publications (8 columns)

Migration: m61_001_stock_intelligence.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.String(128)	None True False
symbol	sa.String(32)	False False False
scanner_run_id	sa.String(128)	False False False
candidate_id	sa.String(128)	True False False
snapshot_timestamp	sa.String(64)	False False False
payload_json	sa.JSON()	False False False

Column	Type	Null / PK / Unique / FK-default metadata
publication_name	sa.String(128)	False False False
status	sa.String(32)	False False False

191. stock_scanner_runs (8 columns)

Migration: m61_001_stock_intelligence.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.String(128)	None True False
symbol	sa.String(32)	False False False
scanner_run_id	sa.String(128)	False False False
candidate_id	sa.String(128)	True False False
snapshot_timestamp	sa.String(64)	False False False
payload_json	sa.JSON()	False False False
status	sa.String(32)	False False False
provider	sa.String(32)	False False False

192. stock_scanner_timeframe_states (9 columns)

Migration: m61_001_stock_intelligence.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.String(128)	None True False
symbol	sa.String(32)	False False False
scanner_run_id	sa.String(128)	False False False
candidate_id	sa.String(128)	True False False
snapshot_timestamp	sa.String(64)	False False False
payload_json	sa.JSON()	False False False
timeframe	sa.String(16)	False False False
direction	sa.String(32)	False False False
structure	sa.String(32)	False False False

193. stock_supply_demand_zones (11 columns)

Migration: m61_002_stock_intelligence.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.String(128)	None True False
symbol	sa.String(32)	False False False
scanner_run_id	sa.String(128)	False False False
candidate_id	sa.String(128)	True False False
snapshot_timestamp	sa.String(64)	False False False
payload_json	sa.JSON()	False False False
timeframe	sa.String(16)	False False False
zone_type	sa.String(32)	False False False
lower_bound	sa.Float	False False False
upper_bound	sa.Float	False False False
strength	sa.Float	False False False

194. stock_support_resistance_levels (10 columns)

Migration: m61_002_stock_intelligence.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.String(128)	None True False
symbol	sa.String(32)	False False False
scanner_run_id	sa.String(128)	False False False
candidate_id	sa.String(128)	True False False
snapshot_timestamp	sa.String(64)	False False False
payload_json	sa.JSON()	False False False
timeframe	sa.String(16)	False False False
level_type	sa.String(32)	False False False
price	sa.Float	False False False
strength	sa.Float	False False False

195. stock_trade_plans (7 columns)

Migration: m61_001_stock_intelligence.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.String(128)	None True False

196. stock_trend_forecast_snapshot (7 columns)

Migration: m52_003_trend_forecasting.py

Column	Type	Null / PK / Unique / FK-default metadata
id	sa.BigInteger()	None True False
symbol	sa.String(32)	False False False
as_of_date	sa.Date()	False False False
snapshot_timestamp	sa.DateTime(timezone=True)	False False False
horizon_days	sa.Integer()	False False False
status	sa.String(32)	False False False
payload_json	postgresql.JSONB(astext_type=sa.Text)	False False

Indexes: op.create_index('ix_trend_forecast_symbol_horizon_timestamp', 'stock_trend_forecast_snapshot', ['symbol', 'horizon_days', 'snapshot_timestamp'])

197. stock_trend_snapshot (18 columns)

Migration: m52_001_trend_intelligence.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_timestamp	sa.DateTime(timezone=True)	False False False
symbol	sa.String(32)	False False False
as_of_date	sa.Date()	False False False
short_term_state	sa.String(40)	False False False
intermediate_term_state	sa.String(40)	False False False
long_term_state	sa.String(40)	False False False
alignment_score	sa.Float()	False False False
trend_quality_score	sa.Float()	False False False
trend_confidence	sa.Float()	False False False
trend_stage	sa.String(40)	False False False
trend_age_days	sa.Integer()	False False False
relative_strength_vs_spy	sa.Float()	False False False

Column	Type	Null / PK / Unique / FK-default metadata
sector	sa.String(100)	False False False
sector_etf	sa.String(20)	False False False
calculation_version	sa.String(40)	False False False
payload_json	sa.JSON()	False False False
created_at	sa.DateTime(timezone=True)	False False False

Indexes: op.create_index('ix_stock_trend_snapshot_symbol_asof', 'stock_trend_snapshot', ['symbol', 'as_of_date'])

198. stock_trend_transition_snapshot (13 columns)

Migration: m52_002_trend_transition_intelligence.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_timestamp	sa.DateTime(timezone=True)	False False False
symbol	sa.String(32)	False False False
as_of_date	sa.Date()	False False False

Indexes: op.create_index('ix_stock_trend_transition_symbol_asof', 'stock_trend_transition_snapshot', ['symbol', 'as_of_date'])

199. symbol_return_snapshot (12 columns)

Migration: m46_001_market_intelligence.py | Model: src/trading_ai/market_intelligence/database_models.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_timestamp	sa.DateTime(timezone=True)	None True False
symbol	sa.String(16)	None True False
as_of_date	sa.Date()	False False False
return_1d	sa.Float()	None False False
return_5d	sa.Float()	None False False
return_20d	sa.Float()	None False False
realized_volatility_20d	sa.Float()	None False False
beta_60d	sa.Float()	None False False
sector	sa.String(96)	None False False
confidence	sa.Float()	False False False
provenance	sa.String(32)	False False False
payload_json	sa.Text()	False False False

200. table (2 columns)

Column	Type	Null / PK / Unique / FK-default metadata
primary_timeframe	sa.String(16)	True False False
contributing_timeframes	postgresql.JSONB(astext_type=sa.Text)	False False

Indexes: op.create_index('ix_{table}_{col}', table, [col])

201. underlying_volatility_snapshot (15 columns)

Migration: m46_002_polygon_authoritative_closure.py

Column	Type	Null / PK / Unique / FK-default metadata
snapshot_timestamp	sa.DateTime(timezone=True)	None True False
underlying_symbol	sa.String(24)	None True False
as_of_date	sa.Date()	False False False
atm_iv_30d	sa.Float()	None False False

Column	Type	Null / PK / Unique / FK-default metadata
realized_vol_20d	sa.Float()	None False False

5. Database governance rules

- Published-current authorities are consumed by explicit publication identity/status rather than arbitrary latest rows.
- Lineage identifiers connect downstream decision/portfolio/execution state back to the scanner/intelligence authority that created it.
- Market-session date and processing timestamp are separate concepts. M77.40 now resolves the effective market date from Stock Scanner lineage rather than portfolio publication timestamp.
- Research/shadow tables do not grant production authority. M78 authority_effect remains false.
- Installer migrations are transactional under PostgreSQL/Alembic and are paired with focused verification and backup/rollback procedures.